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The IUPAC FAIRSpec Metadata Object Model Specification, Version 0.1.3

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1. Introduction

1.1 Audience

The audience for this specification includes:

- Data management professionals, researchers, and publishers who are using IUPAC FAIRSpec Finding Aids and wish to understand them more fully.
- Developers of software such as Electronic Laboratory Notebooks (ELNs), Laboratory Information Management Systems (LIMS), databases, and data repositories who wish to expand their systems to facilitate the findability, accessibility, interoperability, and reusability of their data (whether spectroscopic or not).
- Developers of web applications that will utilize IUPAC FAIRSpec Finding Aids to provide a metadata-rich value-added component to web sites and landing pages for published or local in-house spectroscopic data collections.
- Developers of Application Programming Interfaces (APIs) who want to enhance machine-to-machine communication related to spectroscopic data collections.
- Professionals in the field of chemistry, data science, or computer science who are interested in adding instrumental techniques or other enhancements not already described in the model (which focuses primarily on NMR spectroscopy in its current state, particularly in relation to chemical structures and spectra).
- Anyone from the wider scientific community interested in developing rich discipline-specific metadata models and finding aids themselves.

1.2 The IUPAC FAIRSpec Project

The goal of the [IUPAC Project 2019-031-1-024](#) is to enable a standardized way of managing spectroscopic data digital collections. This document sets forth a set of standards for the description and cataloging of data and its associated metadata in ways that are practical, relatively simple to implement, modular, intuitive, easily extended, and flexible in terms of requirements for the data itself as well as format of the associated metadata.

The scope of the project is *spectroscopic* data within a *chemical* context, but what is described here is fully extensible to any sort of data, within any context, and is already being proposed in the area of materials science.

The proposed standards involve several aspects:

- a set of principles underlying what we mean by "FAIR" in relation to spectroscopic data.
- a detailed **metadata object model** for describing the contents of a "spectroscopic data collection" in terms of objects and relationships of objects as described by metadata,
- a standard for **describing properties** of digital objects within the metadata records of the finding aid,
- a standard for the **serialization of the finding aid** for a collection,
- a recommendation for the **organization** of digital objects within a collection,
- a proposal for and demonstration of methods of **data and metadata extraction** and the generation of IUPAC FAIRSpec Finding Aids.

This document focuses on the second of these bullet points, specifically describing the eleven metadata classes comprising the core of the metadata object model, along with twelve standard digital object-related metadata classes and several additional classes that are suggested for general use and are examples of how the model can be extended to specific applications.

Specifically, this document describes a specification for data management of spectroscopic data in the area of chemistry that allows for a seamless process from generation of experimental data through analysis to publication and public archiving. It describes in depth the abstract model that underlies the metadata associated with an IUPAC FAIRData collection, including the IUPAC FAIRSpec extension of that model. The realization (or "serialization") of the model in the form of *JavaScript Object Notation (JSON)* as an *IUPAC FAIRSpec Finding Aid* describing an *IUPAC FAIRSpec Data Collection* is described.

1.3 IUPAC FAIRSpec Principles

These Principles are elaborated upon in the *Pure and Applied Chemistry* article [IUPAC Specification for the FAIR Management of Spectroscopic Data in Chemistry \(IUPAC FAIRSpec\) - Guiding Principles](#) (accepted Mar 13, 2022). Only a brief summary is given here in order to provide the rationale for the data model.

Spectroscopic data in the area of chemistry are intimately connected to chemical structure. The purpose of carrying out spectroscopic measurements is most often to discover the identity of pure compounds, to identify and quantitate the relative amounts of chemical impurities in a mixture, and to determine the structure of new compounds.

In addition, there are contexts where it is more appropriate to refer to spectroscopic data in relation to "samples." We want to know the identity of a sample, but we don't yet know its chemical structure. Or, perhaps we are working with a material that is not characterizable *per se* as a chemical compound. These standards cover such cases as well.

Spectroscopy datasets can be complex. We make the distinction here between *Digital Entities* (sequences of bytes, for example "files"), which, through metadata association, become *Digital Objects*. [def reference] A key concept in the IUPAC FAIRSpec data model is the **IUPAC FAIRSpec Digital Collection**, which uses metadata to make useful connections between digital objects and allows for a variety of "representations" of those objects (Figure 1).

It is important to understand that the IUPAC FAIRSpec Standard data model is not about standardizing instrument data file formats (though it benefits from that) or requiring specific representations (though it might make specific recommendations in this regard). Rather, it is about providing a baseline common ground for the storage, transmission, and description of the contents of a digital collection derived from spectroscopic measurements or calculations. The ultimate goal is to be able to provide concise descriptions of complex data sets associated with manuscripts, ongoing laboratory work, and teaching, that can be findable, accessible, interoperable, and reusable by machines and humans alike.

One to One and One to Many FAIR Relationships

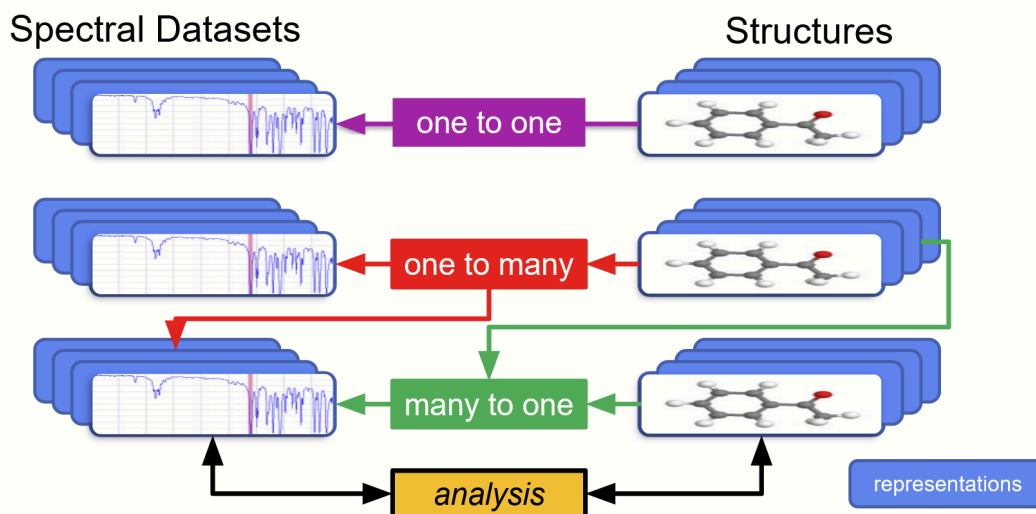


Figure 1. Key relationships among FAIRSpec Digital Objects.

The following principles underlie our development of the IUPAC FAIRSpec standards:

1. FAIR Management of data should be an ongoing concern.

- A. FAIR management of data must be an explicit part of research culture.
- B. FAIR management of data should be of intrinsic value.
- C. Good data management requires distributed curation.
- D. Experimental work is by nature iterative.

2. Context is important.

- A. Digital objects are generally part of a collection.
- B. Chemical properties are related to chemical structure.
- C. Data relationships are diverse and develop over time.
- D. FAIR management of data should allow for validation.

3. FAIR management of data requires curation.

- A. Data reuse relies upon practical findability.
- B. Data has to be organized to be accessible.
- C. Data interoperability requires well-designed metadata.
- D. Value is in the eye of the reuser.

4. Metadata must be standardized and registered.

- A. Register key metadata.
- B. Assign a variety of persistent identifiers.
- C. Enable metadata crosswalks.
- D. Allow for value-added benefits.

5. FAIR data management standards should be modular, extensible, and flexible

- A. Modularity allows specialization.
- B. Allow for future needs.
- C. Respect format and implementation diversity.
- D. All data formats should be valued.

Specifically in the context of chemistry and spectroscopy, these five main principles can be interpreted as below:

1. FAIR Management of data should be an ongoing concern.

- Don't wait until publication time to organize your data.
- Recognize the ongoing value of well-organized data.
- Allow for corrections and addition of new information.

2. Context is important.

- Associate spectra with chemical structure as much as possible.
- Allow for ambiguity and the reconsideration of these associations.
- Find ways to validate your structural and spectral analysis.

3. FAIR management of data requires curation.

- Accept that you are going to have to do part of the work.
- Optimize opportunities for data citation.
- Do not presume to know how people will utilize your data.

4. Metadata must be registered and standardized.

- Findability relies upon proper registration.
- Work with data management professionals in your organization.
- Include discipline-specific metadata.

5. FAIR data management standards should be modular, extensible, and flexible.

- FAIR data management should be as simple as possible.
- Find (or create!) the right tools for the job.
- Find ways to make data management useful to you and your project *now*.

As outlined in Figure 2, the emphasis here is on *workflow*. From the first moment that a spectroscopic dataset is produced by an instrument, best-practices data management can start to prepare a dataset for future publication in association with a scholarly work. Development of a *metadata model* provides a basis for this workflow. The IUPAC FAIRSpec Finding Aid is simply a concretization of this model. Thus

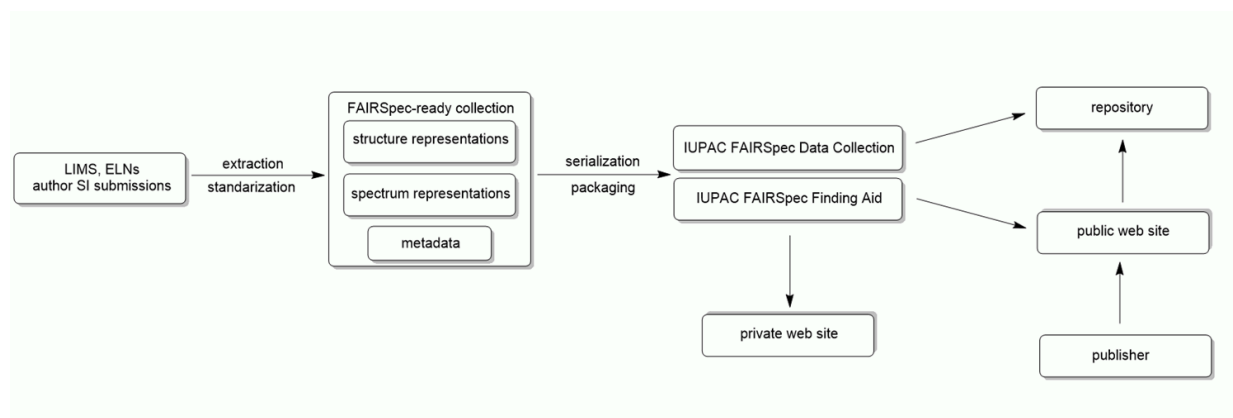


Figure 2. A data management workflow enabled by the production of an IUPAC FAIRSpec Data Collection and its accompanying IUPAC FAIRSpec Finding Aid.

1.4 Resources for Implementers

In association with the development of the metadata model, we have produced a full implementation of the model using the *Java* programming language. It can be found at the GitHub repository for the FAIRSpec project at <https://github.com/IUPAC/IUPAC-FAIRSpec>. This implementation includes an "extractor" program that can automatically or semiautomatically extract metadata from "FAIRSpec-ready" data collections, flag and possibly correct issues relating to compound structure and spectroscopic issues. The output of the extractor includes an IUPAC FAIRSpec Finding Aid, an IUPAC FAIRSpec Data Collection (only if desired), and numerous log and warning message files. A sample "landing page" is produced that can be used to display a web-based visualization of the data collection, summarize its contents, assist in verifying the integrity of the collection, and prepare the data collection for publication or submission to a repository. The extractor program is not meant to be a finished product of the IUPAC project. Rather, it simply expresses what is stated in this metadata model discussed here in concrete terms.

The GitHub site includes a wide-ranging set of sample FAIRSpec datasets with associated IUPAC FAIRSpec Finding Aids can be found at <https://iupac.github.io/IUPAC-FAIRSpec>. A presentation of the principles associated with this work are elaborated upon in the *Pure and Applied Chemistry* article [IUPAC Specification for the FAIR Management of Spectroscopic Data in Chemistry \(IUPAC FAIRSpec\) - Guiding Principles](#) (2022). An IUPAC Technical Report, [FAIRSpec-Ready Spectroscopic Data Collections – Advice for Researchers, Authors, and Data Managers](#) (2025) describes how researchers and authors can prepare spectroscopic data collections to be ready for automated processing and the preparation of IUPAC FAIRSpec Finding Aids.

2. The IUPAC FAIRData Metadata Object Model

What follows is a general discussion of the metadata object model underlying the IUPAC FAIRSpec specification. We refer to this model as the "IUPAC FAIRData" model because it is a general model that is not specific in any way to spectroscopy. "FAIRSpec" refers to the IUPAC FAIRSpec extension of the IUPAC FAIRData Object Model. The point here being that the model is easily extensible, and IUPAC FAIRSpec is just one example of how that can be accomplished.

2.1 Definitions

Although this discussion is framed in the language of object-oriented programming in order to more clearly define its relational aspects, scope, and limitations, it is designed for readers not familiar with programming. To this end, we offer a few definitions that will be useful for programmers and nonprogrammers alike. These include:

- *object*

A construct having properties (also called *fields*) that describe it or its relationships to other objects. An example of an object might be a structure or a spectrum. In this context, objects may be found alone (the finding aid itself) or as a collection objects,

which itself is an object. An object may be *representable* in the form of bits and bytes (a *digital object*) or it may be just an abstract set of properties (a *metadata object*).

- *class*

An abstract description of an object's type, expressing its capabilities and limitations. We say that an object is an "instance of its class", or that "instantiation of a class creates an object". The class defines what properties its objects are allowed to have, and the object defines those properties. In this standard, the names of all IUPAC FAIRData Metadata Object Model classes start with "IFD". IUPAC FAIRSpec extensions of the model (Section 3) start with "FAIRSpec".

- *subclass and superclass*

Classes are part of a hierarchy including *superclasses* and *subclasses*. Thus, the "child" of a superclass is a subclass, and the "parent" of a subclass is a superclass. We say that a subclass *extends* its superclass so that it can "inherit" the superclass's properties. This efficient design allows general characteristics to be described in one class (the superclass) and automatically propagated to a whole family of other classes (its subclasses). Subclassing allows for expansion and customization of the model without the need to change the core classes. So, for example, we will see that the key aspect of an *IFDCollection* is that it is a list of IFDObjects. This is also true of its subclass *IFDStructureCollection*, but the *IFDStructureCollection* class is more restrictive. It must be a list of only specific types of IFDObject, specifically *IFDStructure* objects.

- *abstract class*

An abstract class is a class that can only be a superclass and can never be instantiated itself. It is basically a template that provides common properties for its various subclasses. Declaring a class to be abstract indicates that it will be used in a variety of manners, as specified by its different subclasses. The class *IFDRepresentation* is an example of an abstract class, defining general properties of digital objects – objects that can be represented in the form of bytes or characters (often referred to as "files"). Subclasses include *IFDStructureRepresentation*, *IFDDataObjectRepresentation*, *IFDSampleRepresentation*, and *IFDAnalysisObjectRepresentation*. The FAIRSpec extension further subclasses *IFDDataObjectRepresentation* to be specific for a given technique — for example, as *FAIRSpecNMRRepresentation*.

- *field*

In our context, a *field* is an aspect of a class that has a data type, such as "string", "integer", "float", "array", or "map". Data types can be qualified. For example, we can have "an array of strings" or "a map of strings to other strings." Examples of fields include *id*, *description*, and *label*. The primary data types utilized in this metadata model are described in Appendix A.

- *serialization*

Metadata model *serialization* is the process of "rendering" a set of metadata classes in the form of a digital object, as bytes. The form of the serialization is the decision of

the implementer. The prototype implementation we have developed for the FAIRSpec Project and that is used for examples here is *JavaScript Object Notation (JSON)*. But it is also possible that the model might be implemented in a mark-up language such as XML or HTML. It is important to understand that a metadata model stands on its own, as a concept independent of its serialization.

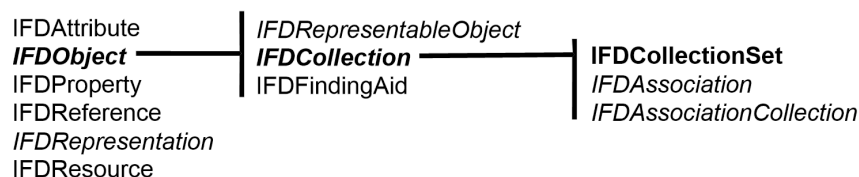
2.2 Model classes

2.2.1 Overview

An overview of the classes in the IUPAC FAIRData Metadata Model is shown in Figure 3. The model includes two primary groups of classes: core classes and specialized subclasses. Core classes are completely general, applicable to any metadata model. Specialized subclasses add concepts that are associated with experimental datasets, such as sample, structure, data, and analysis. Each of these classes is described more formally in Appendix B.

IUPAC FAIRData Metadata Object Model classes

Core Classes



Specialized subclasses

IFDRepresentableObject

IFDSample
IFDStructure
IFDDataObject
IFDAnalysisObject

IFDCollection

IFDSampleCollection
IFDStructureCollection
IFDDataObjectCollection
IFDAnalysisObjectCollection

IFDRepresentation

IFDSampleRepresentation
IFDStructureRepresentation
IFDDataObjectRepresentation
IFDAnalysisObjectRepresentation

IFDAssociation

IFDSampleDataAssociation
IFDSampleStructureAssociation
IFDStructureDataAssociation
IFDStructureDataAnalysisAssociation

IFDAssociationCollection

IFDSampleDataAssociationCollection
IFDSampleStructureAssociationCollection
IFDStructureDataAssociationCollection
IFDStructureDataAnalysisAssociationCollection

Figure 3. Main classes of the IUPAC FAIRData Metadata Object Model. Abstract classes (classes that can only serve as superclasses) are *italicized*. The classes can be subdivided into two types: core and derived. [MOM-coreClasses.cdxml]

2.2.2 Core classes

Twelve core classes define the most basic elements of the IUPAC FAIRData Metadata Model, as depicted in the first section of Figure 3.

2.2.2.1 IFDObject and its core subclasses

IFDObject is the most basic object in the model. It is simply a "list of things", specifically other IFDObjects and their associated metadata. The function of IFDObject is to provide core properties that are common to all objects, including the collection itself. These include:

- *id*, a unique identifier for the object within its collection, such as "1a".
- *label*, text that can be used as a label for the object, but not necessarily unique.
- *description*, any descriptive, possibly searchable, text associated with the object
- *note*, generally a production note, perhaps indicating issues with this object
- *ifdProperties*, a set of metadata key:value pairs that are described by the standard; for example, IFD.property.dataobject.fairspec.nmr.solvent_InChI.
- *attributes*, additional metadata key:value pairs not described by the standard, such as "solvent". Attributes allow implementers to introduce properties that are important to their work without breaking the standard. Future versions of the standard may include those properties using standardized syntax, adding requirements relating to units and data type.

IFDObject is the immediate superclass of the key classes, **IFDRepresentableObject**, **IFDCollection**, and **IFDFindingAid**.

The **IFDRepresentableObject** class correlates to an object in the model that can be represented in one or more ways in the form of bytes or characters. The class is abstract because, in order to give it substance, we need to know *what kind* of object we are talking about. (More on this in Section 3.) This class maintains a list of **IFDRepresentation** instances, all characterizing the same object. For example, given a "chemical structure" or an "NMR spectral dataset" (the object), we could have any number of distinct representations — an image of a spectrum, a PDF document, a number of structure files with a specific format, such as 2D or 3D MOL, CDXML, or CIF. The IFDRepresentableObject collects these various representations of the same object.

The **IFDCollection** class is a widely used superclass providing a listing of *specific kinds* of IFDObjects. Thus, while IFDObject is already a list and maintains metadata about that list, IFDCollection creates a template for working with specialized sorts of lists. For example, subclasses of IFDCollection maintain metadata relating specifically to samples. Thus, IFDStructure is an IFDCollection of IFDRepresentations related to a specific chemical structure, and IFDStructureCollection is an IFDCollection of IFDStructures. IFDCollectionSet is the principal IFDCollection in an IUPAC FAIRData Collection.

The **IFDFindingAid** class is the "top-level" class, maintaining metadata relating to the collection as a whole, such as a list of identified resources, publication and repository information, and licensing

information. It contains a single IFDCollectionSet. It is the serialization of the IFDFindingAid class that creates the human- and machine-readable digital document we refer to as the IUPAC FAIRSpec Finding Aid.

2.2.2.2 Core subclasses of IFDCollection

The three core subclasses of IFDCollection include **IFDCollectionSet**, **IFDAssociation**, and **IFDAssociationCollection**.

The **IFDCollectionSet** class maintains the primary metadata collections for a IUPAC FAIRData Collection. These include sample and/or structure collections, a dataset collection, and collections that associate items in these various collections to each other.

One of the most important subclasses of IFDCollection is **IFDAssociation**. This class maintains a "collection of collections". It is the place where the "one-to-many" or "many-to-many" or "many-to-one" relationships depicted in Figure 1 are found. The FAIRSpec extension implements a very important subclass of IFDAssociation, namely **FAIRSpecCompoundAssociation** (Section 3.4). Each of these associations represents a distinct "chemical compound", which in this context is itself a collection that relates some number of chemical structures to some number of instrumental datasets.

The IFDCollectionSet maintains one or more collections of IFDAssociations as **IFDAssociationCollection** objects. For example, in a FAIRSpec Finding Aid related to a publication, we expect to find an IFDAssociationCollection named "compounds" that relates chemical structures to some set of spectra, here serialized as JSON:

```
"compounds":{
  "itemsByID": {
    "3aa" : {
      "itemsByID": {
        "structures": ["3aa"],
        "spectra":["3aa-C","3aa-H"]
      }
    },
    ...
  }
  ...
}
```

This says that Compound 3aa has a single chemical structure associated with it, namely Structure 3aa. Two spectra are associated with this structure, namely Spectra 3aa-C and 3aa-H. Note that each of these structures and spectra are themselves collections of representations (SMILES strings, MOL files, PNG images, data files, etc.). Associations essentially provide the "edges" to an abstract graph having IFDRepresentableObject nodes. The *compounds* collection as a whole connects all of the spectra in an IUPAC FAIRSpec Data Collection with their associated chemical structures.

2.2.2.3 Additional core classes

Besides IFDObject, core classes providing general metadata include **IFDRepresentation**, **IFDAttribute**, **IFDProperty**, **IFDReference**, and **IFDResource**, as depicted also in Figure 3.

The **IFDRepresentation** class maintains descriptive metadata relating to specific digital objects in an IUPAC FAIRData Collection. That said, an IFDRepresentation may or may not refer to an actual digital object ("file") in the collection. A metadata extractor with linked chemistry-related libraries, for example, might create a variety of representations during the process of extraction as "value-added" additions to the collection. For example, given a CDXML file or even just a SMILES string, the extractor might create a 2D MOL file, a 3D MOL file, an InChI, and an InChIKey – all of which are IFDRepresentations – and simply store these in the finding aid itself rather than creating new files for each of them. (In fact, *any* digital object in a collection in principle can be represented as a string within the finding aid itself, as the standard allows any sequence of bytes to be represented as a base64-encoded string.) An extractor with links to web-based services could provide all of these representations along with predicted NMR spectra, validation reports, and proposed detailed structure-spectrum analyses.

IFDProperty and **IFDAttribute** are storage classes for key:value pairs. **IFDProperty** is reserved for standardized metadata key/value pairs for all IFDObjects. The keys of an IFDProperty all start with "IFD.property." For example, *IFD.property.structure.inchikey* or *IFD.property.dataobject.expt_title*. IFDProperties may have specified data types (String or Number), value ranges ([1...]), value formats (1H, 13C, ...), or units (e.g., Hz, MHz, Kelvin, ...) that may be specified in this standard or its extension. If such specifications are made, they should be declared in the formal specifications of the serialization, not in the finding aid itself. **IFDAttribute** is the storage class for *non-standardized* metadata key/value pairs for all IFDObjects. An IFDAttribute can use one name to refer to a single value or an array of values. Its allowed data types are String, Number, Boolean, and Array. A Finding Aid generator may create any number of relevant IFDAttributes for any IFDObject.

IFDResource is a small general-purpose structure for holding the byte length and reference for an external resource digital object from which the Finding Aid was created. For example, the finding aid may point to a dataset that is in a university repository in the form of a ZIP file. If an IUPAC FAIRData Collection has been created, an IFDResource will be created for it and will be stored in the *resources* field of an IFDFindingAid.

IFDReference is the storage class connecting IFDRepresentations to their original source digital objects within an IUPAC FAIRData Collection.

2.2.3 Specialized subclasses

These classes, depicted in the lower part of Figure 3, relate more specifically to chemical data. In this version of the IUPAC FAIRData Metadata Model, these involve samples, structures, data objects, and analysis objects. An IFDCollectionSet is simply a collection of collections, maintaining zero or one each of a sample collection, a structure collection, a collection of data objects, and a collection of analysis objects. In addition – and key to the model's design – the IFDCollectionSet maintains one or more *associative* collections that maintains a list of specific associations of these four types of object. The most common of these is the structure-data object collection, which

maintains a list of structure-data object associations. These associations ("compounds" in the context of FAIRSpec) relate one or more chemical structures (each with one or more representations) to one or more instrument data sets ("spectra" in the context of FAIRSpec, each also with one or more representations).

2.2.3.1 *IFDRepresentableObject* and *IFDCollection<IFDRepresentableObject>*

- **IFDSample** A class for metadata relating to a physical sample, typically with at least an id field.
- **IFDStructure** A class to maintain metadata relating to a single chemical structure
- **IFDDataObject** An abstract class used to distinguish metadata relating to representations of

The IUPAC FAIRData Metadata Model includes four extensions to *IFDRepresentableObject* and their respective collections. These include: a specific experimental, simulated, or predicted instrument dataset.

- **IFDAnalysisObject** An abstract class identifying a generally detailed analysis of the relationship of one or more chemical structures with one or more experimental datasets. Thus, "analysis" in this context is more specific than the general sense of "chemical analysis". Typically this object's representations relate to the interpretation of instrumental data in terms of molecular structures. For example, the FAIRSpec extension subclasses *IFDAnalysisObject* as **FAIRSpecNMRAnalysisObject** and associates one or more specific structure representations (a specifically numbered MOL file, perhaps) with one or more NMR spectra or peak listings, along with representations such as assignment tables, NOE and coupling constant analyses, and detailed peak assignments.

2.2.3.2 *IFDAssociation* and *IFDAssociationCollection*

One of the primary features of an IUPAC FAIRSpec Data Collection is that it can express *associations* among representable objects. Four such associations are defined in this specification because they are ubiquitous; extensions may add more. These include:

- **IFDSampleDataAssociation** An association that relates a *specific* physical sample to an experimental dataset, the *IFDSampleDataAssociation* may be particularly useful early on in the laboratory workflow, when an instrument dataset is created for a sample without foreknowledge of chemical structure. The sample object may simply declare an id, but if that is the case, an alternative is simply to associate an instrument datasets with samples using the *IFDDataObject*.*originating_sample_id* field. In other cases, where additional metadata is involved, this association allows associating all that sample metadata with one or more instrument datasets.
- **IFDSampleStructureAssociation** If a sample database is available, perhaps associated with an ELN[[def]], the model may include a sample-structure association (one-to-many and many-to-many). The sample-structure association is designed to be used only when there is no available experimental data linking a sample with a chemical structure. This might be the case, for example, when a commercial sample is used and there is no need to verify its chemical nature via spectroscopy.

- IFDStructureDataAssociation** The most important association in an IUPAC FAIRData Collection is the structure-data object association. Without this association, there is little that can be made of a collection of instrument datasets. The connection between structure and spectrum is important enough that the FAIRSpec extension subclasses this class as a **FAIRSpecCompoundAssociation**. That is to say, in chemistry one of the two most important aspects of a "chemical compound" is its chemical structure and its spectra. We often say that *this* spectrum is of *this* compound, which has *this* structure.
- IFDStructureDataAnalysisAssociation** One of the principal "proof of structure" for a compound in a chemical laboratory is an analysis of its spectra. This class is intended to represent a detailed correlation between chemical structure for a compound and its related experimental and/or predicted spectroscopic data. It contains three collections, an IFDStructureCollection, an IFDDataObjectCollection, and an IFDAnalysisObjectCollection. For instance, the association might correlate specific atoms or groups of atoms of a (specific) chemical structure representation with specific signals in a spectrum or other sort of data object. Generally there will be only one structure with one representation in this association, in order to identify specific atoms in the structure by name or number.

2.2.3.3 IFDRepresentation subclasses

Each of the *IFDRepresentableObject* subclasses is itself a collection of *IFDRepresentations*. Thus, we have representations of a chemical sample (IFDSampleRepresentation), a chemical structure (IFDStructureRepresentation), a data object (IFDDataObjectRepresentation), and an analysis object (IFDAnalysisObjectRepresentation). Examples of each of these representation types are given in Table 1.

Table 1. Examples of IFDRepresentation types.	
Representation Type	Examples
sample	image of a crystal
structure	MOL or CDXML file, SMILES or InChI, 2D drawing, PNG image of 3D model
data object	ZIP file or file directory produced by an NMR instrument, peak listing, JCAMP-DX file, predicted spectrum image, PDF document containing a spectrum, third-party post-processing document
analysis object	table of peak assignments and coupling constants in NMR, assigned vibrations in IR

2.3 Extending the model – contributed subclasses and metadata specifics

Importantly, the model is designed to allow for customization in ways that go well beyond generic data. Any of the classes of the model can be subclassed further to provide a variety of additional, even more specialized object types, allowing for collections of different sorts. All that is required by the model is that the model classes themselves are still identifiable, and that the subclassing is well described. Customization can be accomplished in any combination of the following three ways:

Contributed subclasses can be designed and added to the model. Specifically, the **FAIRSpecFindingAid** class is a subclass of *IFDFindingAid*. It describes a **FAIRSpecCollection**, which is a subclass of *IFDCollectionSet*. The process of subclassing introduces specialized metadata properties and representations that are specific in this case to the area of spectroscopy involved. The FAIRSpecFindingAid class can be serialized (rendered in bytes) in a machine-readable form, such as JSON or XML. , of that produces the ultimate digital object we refer to as the *IUPAC FAIRSpec Finding Aid*, which describes the contents of the actual *IUPAC FAIRSpec Data Collection*.

Elaboration of metadata elements provides another means of customization. Note that the model itself is largely independent of the details and format of the descriptive metadata key/value properties themselves that we traditionally associate with "metadata". This is by design. Though there are a few basic rules in the model as to how these keys in particular are constructed (see [ifd.properties](#)), the details of those keys and their associated values are left to extension contributors.

Addition of nonstandard attributes is also allowed. Nothing in the specification requires the *exclusive* use of standardized IUPAC FAIRData metadata key/value pairs. The finding aid has two property-like categories: "*ifdProperties*" are metadata elements that have standardized IUPAC FAIRData (or its extensions') keys and allowed values; "*attributes*" are metadata of completely free format, or of a format described specifically by the implementor or by other standards.

3 The FAIRSpec Extension

3.1 Overview

The FAIRSpec extension (org.iupac.fairdata.contrib.fairspec) supplies an additional spectroscopy-specific set of subclasses and additional IFDProperty key/value pairs to the IFDFAIRData classes. These additions are diagrammed in Figure 4 and described in detail in Appendix C.

IUPAC FAIRSpec extension subclasses

FAIRSpec subclasses	Superclass
FAIRSpecFindingAid	IFDFindingAid
FAIRSpecCollection	IFDCollectionSet
<i>FAIRSpecDataObject</i>	<i>IFDDataObject</i>
<i>FAIRSpecAnalysisObject</i>	<i>IFDAnalysisObject</i>
FAIRSpecCompoundAssociation	IFDStructureDataAssociation
FAIRSpecCompoundCollection	IFDStructureDataAssociationCollection
<i>FAIRSpecAnalysisAssociation</i>	IFDStructureDataAnalysisAssociation
<i>FAIRSpecAnalysisCollection</i>	IFDStructureDataAnalysisAssociationCollection
IFD classes used without subclassing	
	IFDSample
	IFDSampleCollection
	IFDStructure
	IFDStructureCollection
	IFDSampleDataAssociation
	IFDSampleDataAssociationCollection

Figure 4: Primary FAIRSpec subclasses with their associated IUPAC FAIRData superclass.

3.2 FAIRSpecFindingAid and FAIRSpecCollection

The **FAIRSpecFindingAid** subclass adds a "brand" to a collection by subclassing IFDFindingAid. When serialized, an object of the FAIRSpecFindingAid class becomes the IUPAC FAIRSpec Finding Aid digital object that ultimately distinguishes an IUPAC FAIRSpec Data Collection from other sorts of digital collections. The JSON serialization of a FAIRSpecFindingAid class provides a readily accessible representation of the entire data collection. When integrated into a web-based landing page this JSON document, the IUPAC FAIRSpec Finding Aid, can serve as the target of a persistent identifier such as a DOI (Digital Object Identifier) for the entire data collection. In the context of an ongoing research program, the IUPAC FAIRSpec Finding Aid can serve as a readily available index of relevant spectral analyzes related to a specific researcher, subgroup, or research group. In the

context of a searchable database, the IUPAC FAIRSpec Finding Aid serialized as JSON could be dynamically created in response to an API query, allowing internationally standardized programmatic access to the collection.

The **FAIRSpecCollection** class subclasses IFDCollectionSet, specifically declaring (in its current version) five specific IFDCollection<IFDRepresentableObject> objects (read "IFDCollection of IFDRepresentableObject objects"):

- "samples" (an IFDSampleCollection),
- "structures" (an IFDStructureCollection),
- "spectra" (an IFDDataObjectCollection),
- "compounds" (an IFDAssociationCollection), and
- "analyses" (an IFDStructureDataAssociationCollection).

3.3 FAIRSpecDataObject and Instrumental technique-specific subclasses

The **FAIRSpecDataObject** is an abstract class that serves as the superclass identifying its subclasses as being spectroscopy related. As a starting point, this version of the model implements the following subclasses of *FAIRSpecDataObject*:

FAIRSpecCOMPData
FAIRSpecIRData
FAIRSpecNMRData
FAIRSpecMSData
FAIRSpecHRMSData
FAIRSpecRamanData
FAIRSpecUVVISData

Clearly this list could be extended further; future versions of this specification may include additional spectroscopic methods.

3.4 FAIRSpecCompoundAssociation and FAIRSpecCompoundCollection classes

FAIRSpecCompoundAssociation represents an abstract association specifically involving one or more structures with zero or more spectra. The FAIRSpecCompoundAssociation is like all IFDAssociations in that it is a purely metadata object that has no direct representations as such. It provides the ability to associate multiple structures (isomers of "Compound 3a", for example) with multiple spectra (1H NMR, 13C NMR, etc.). It allows for use of the term "compound" in its less restrictive common use in the literature as "one or more isomerically related compounds"). It could have a property that expresses this relationship, such as a mInChI, but this is not established in this version.

If there are no spectra, then the FAIRSpecCompoundAssociation becomes simply an association of isomerically related structures, much the same way a PubChem "compound" CID 21508 (2,3-dibromobutane) [<https://pubchem.ncbi.nlm.nih.gov/compound/2,3-dibromoButane>] is

ambiguous, describing any one of four stereoisomers, or some mixture of them. However, the FAIRSpecCompoundAssociation contains an IFDStructure for each component, including one or more representations for each, individually. Additional context can be provided by the addition of properties or attributes that express the relationship and relative amounts of the related structures within the association.

In the case where the FAIRSpecCompoundAssociation includes one or more associated spectra, a single IFDCompoundAssociation can express a mixture of constitutionally unrelated chemical compounds that are associated with a set of spectra, "Compounds 3 and 4", for example, where 3 was the reactant and 4 was the product in a chemical reaction, and both are present in an NMR sample. In this case, the structures would have different properties or representations that unambiguously identify them as distinctly different chemical compounds. Future versions of the specification could add a property to FAIRSpecCompoundAssociation that would allow a quantitative description of the associated chemical compounds along the lines of a mInChI (mixture InChI).

The **FAIRSpecCompoundCollection** is a collection of FAIRSpecCompoundAssociations and forms a key element of an IUPAC FAIRSpec Data Collection, along with an IFDStructureCollection and an IFDDataObjectCollection.

3.5 Instrumental technique-specific properties

The IUPAC FAIRSpec extension adds a finite list of metadata key/value pairs specific to particular instrumental techniques. These definitions will evolve as the standard develops. Current definitions can be found in the [IFD.properties file](#) at the IUPAC GitHub site. As of this version, for example, NMR properties include:

IFD.property.dataobject.fairspec.nmr.expt_dim
IFD.property.dataobject.fairspec.nmr.expt_offset_freq1
IFD.property.dataobject.fairspec.nmr.expt_offset_freq2
IFD.property.dataobject.fairspec.nmr.expt_offset_freq3
IFD.property.dataobject.fairspec.nmr.expt_nucl1
IFD.property.dataobject.fairspec.nmr.expt_nucl2
IFD.property.dataobject.fairspec.nmr.expt_nucl3
IFD.property.dataobject.fairspec.nmr.expt_pulse_prog
IFD.property.dataobject.fairspec.nmr.expt_solvent
IFD.property.dataobject.fairspec.nmr.expt_solvent_InChI
IFD.property.dataobject.fairspec.nmr.expt_solvent_InChIKey
IFD.property.dataobject.fairspec.nmr.expt_solvent_common_name
IFD.property.dataobject.fairspec.nmr.expt_thermodynamic_temperature
IFD.property.dataobject.fairspec.nmr.expt_title
IFD.property.dataobject.fairspec.nmr.instr_manufacturer_name
IFD.property.dataobject.fairspec.nmr.instr_probe_type
IFD.property.dataobject.fairspec.nmr.instr_proton_freq
IFD.property.dataobject.fairspec.nmr.instr_nominal_freq

4 Relation of the metadata model to the finding aid

5 Summary

Appendix A. Recognized IUPAC FAIRData Data Types

Data Type	Java Equivalent	Python Equivalent	C++ Equivalent
Byte	byte	byte	char
Integer	int or long*	int	int
Double	double	float	double
Boolean	boolean	bool	bool
String	String (UTF-8)	string (UTF-8)	std::string
byte	byte	byte	char
array	List or array*	list	array
Map	Map	dict	std::map
Date**	String	string	std::string

*The model does not make a distinction between specific implementations of data types such as int/long or List/array. In this specification, we will use them synonymously.

** Dates are Strings in the ISO-8601[<https://www.iso.org/iso-8601-date-and-time-format.html>] format (for example, 2019-05-09T17:15:35-07:00).

Appendix B. Core Classes

2.3.1 IFDObject (abstract)

IFDObject is an abstract superclass for nearly all IUPAC FAIRData Metadata Object Model classes.

- IFDObject

- IFDCollection
 - IFDAssociation
 - IFDAssociationCollection
 - IFDCollectionSet
 - IFDSampleCollection
 - IFDStructureCollection
 - ...(several more)
- IFDRepresentableObject
 - IFDAnalysisObject
 - IFDDataObject
 - IFDSample
 - IFDStructure
- IFDFindingAid

Figure xxx. Subclasses of IFDObject in the IUPAC FAIRData Metadata Object Model.

org.iupac.fairdata.core.IFDObject		
field name	data type	description
description	String	an optional
doi	String	the full DOI reference to this digital object, including the protocol and host (for example, https://doi.org/10.14469/hpc/11741); generally to an HTML landing page describing this object – for example, to a repository page for a collection. For an IFDRepresentableObject, each of the object's representations might have its own url. (See IFDReference .)
id	String	an optional unique identifier for this object; if present, must be unique within this finding aid
ifdType	String	required; the class name of this object, such as <i>org.iupac.fairdata.sample.IFDSample</i>
ifdTypeExtends	String	required for non-core classes; a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to the first class name that is of the form <i>org.iupac.fairdata.core.*</i> (only present for non-core classes)
label	String	an optional short descriptive but not necessarily unique label for this object
note	String	an optional free text note, possibly relating to issues in the production of the finding aid or collection; not intended for general metadata indexing
attributes	Map<String, String>	an optional set of key/value pairs that are not part of the IUPAC FAIRData standard or its extension.

		Characters of the keys must be one of [A-Za-z0-9_] or the space character and must be trimmed of any space character.
ifdProperties	Map<String, Object>	an optional set of key/value pairs where the key is a fully qualified IUPAC FAIRData property name starting with "IFD." or just its associated short (underscore-separated) name, if the propertyPrefix field is also present. The value must be of the form specified by this standard or its extension
propertyPrefix	String	optional; a prefix starting with "IFD." that should be prepended along with a "." to each property key not already starting with "IFD." in order to produce a fully qualified IUPAC FAIRData property name as specified by this standard or its extension
url	String	a URL string targeting this object's sole digital item. The URL field is only present if the doi field is empty and should not contain a DOI itself. For an IFDRepresentableObject, each of the object's representations might have its own url. (See IFDReference .)

Note that properties and attributes both declare metadata properties as key:value pairs. The properties field is reserved for properties that are described by the standard, while the attributes field holds properties that are not (yet) part of the standard. This allows implementers to introduce properties that are important to their work without breaking the standard. Future versions of the standard may include those properties using standardized syntax, adding information about units and type. In addition, because IFDObject extends ArrayList (a Java class maintaining an array of objects that can be added to dynamically), all IFDObjects are inherently list-like.

IFDObject has three direct subclasses: IFDRepresentableObject, IFDCollection, and IFDFindingAid. An IFDRepresentableObject is a list of IFDRepresentation items, while an IFDCollection is a list of IFDRepresentableObject items. The IFDFindingAid is a special object that is not itself a collection but holds the IFDCollectionSet. These three subclasses are described in detail in section 2.3 *Core Classes*.

[2.3.2 IFDRepresentation \(abstract\)](#)

IFDRepresentation is an abstract class with instances that either reference actual digital objects or contain byte sequence or string data themselves. These may be images, spreadsheets, complex analyses, discipline-specific or vendor-specific file formats – any digital format that data (in the broad sense, including chemical structure "data") can be represented. All actual digital objects in an IUPAC FAIRData Collection will be represented in the model as an IFDRepresentation of one type or another and serialized as part of a List<IFDRepresentation>, where each representation in the list is in the form of a Map<String, String>. Note that IFDRepresentation objects themselves are not

IFDObject objects and are not list-like. As an abstract class, IFDRepresentation can only be "realized" as a more specific sort of representation, for example:

- IFDRepresentation
 - IFDAnalysisObjectRepresentation
 - IFDDataObjectRepresentation
 - IFDSampleRepresentation
 - IFDStructureRepresentation

Figure xxx. Subclassing IFDRepresentation differentiates distinctly different types of representations.

Subclassing IFDRepresentation provides a way of ensuring that the correct general type of digital object in an IUPAC FAIRData Collection corresponds with a given IFDRepresentableObject. For example, chemical structure-related metadata objects (IFDStructure objects) must be lists of structure representations (IFDStructureRepresentation objects, such as MOL files). We say that IFDStructure (which is a List) extends IFDRepresentableObject<IFDStructureRepresentation>.

org.iupac.fairdata.core.IFDRepresentation		
field name	data type	description
data	String or byte[]	When the representation itself is suitably small in terms of bytes (in the opinion of the implementer), the representation can be serialized directly within the finding aid as a simple String when appropriate, for example, a SMILES string or InChI key. Alternatively, the value can take the form of a byte array and serialized as a base64-encoded String having the eight-character prefix ;base64,. Base64 encoding is described in RFC 4648. If the <i>data</i> field is not present or null, then the <i>ref</i> field must indicate an originPath or a localPath or a localName. If the <i>data</i> field is present and not null, then the <i>ref</i> field may or may not contain these references. For example, an InChI generated by a metadata extractor from a CDXML file will not have any local or origin path. An image stored within a Finding Aid may still have all of these.
ifdType	String	the object type of the form <i>IFD.representation...</i> declared in the extension of the specification
key	String	the name of this representation, starting with "IFD.representation." and continuing with a dot-separated set of qualifiers that consist only of lower-case or digit [a-z0-9] characters; for example: <i>IFD.representation.dataobject.fairspec.nmr.image</i>
len	Integer	the byte length of the digital object associated with

		this representation
mediaType	String	the media type [rfc....] of this digital object
ref	IFDReference	the machine-followable reference to the digital object associated with this reference, or <i>null</i> (without quotes) or absent when there is no such reference and the <i>data</i> field is non-null.

2.3.3 IFDProperty

IFDProperty is the storage class for standardized metadata key/value pairs for all IFDObjects. This class can be serialized in-line as a key:value pair within a Map<String,String> serialization.

org.iupac.fairdata.core.IFDProperty		
field name	data type	description
(key)	String	The name of this property, starting with "IFD.property." and continuing with a dot-separated set of qualifiers that consist only of lower-case or digit [a-z0-9] characters; for example: <i>IFD.property.dataobject.spec.nmr.expt.offset_freq1</i>
(value)	(variable)	the value of this metadata object, serialized as the data type and declared by the standard or its extension, in the units specified by the standard or its extension

Properties, in principle, may have specified data types (String, Integer, Integer, Float64, ...), units (Hz, MHz, Kelvin, ...), values, value ranges ([1...]), or value formats (1H, 13C, ...) that are specified in the standard or its extension. Properties may be serialized as key:value pairs within a map (as in JSON) or as attribute declarations such as *key=value* (as in XML).

2.3.3 IFDAttribute

IFDAttribute is the storage class for non-standardized metadata key/value pairs for all IFDObjects. An IFDAttributes can use one name to refer to a single value or a list of values.

org.iupac.fairdata.core.IFDProperty		
field name	data type	description
name	String	The name of this attribute consisting of a dot-separated sequence referring to a class followed by a dot and a sequence of lower-case or digit or underscore [a-z0-9] characters; for example: <i>IFD.property.collectionset.source_data_uri</i>
value	(variable)	the value of this metadata object, serialized as a number, boolean, string, or a list of such values.

2.3.5 IFDCollection

IFDCollection is an abstract class designed to hold a set of objects of the same class or superclass. IFDCollection objects are not themselves IFDRepresentableObjects and thus have no digital representation themselves. Note that IFDCollection objects may be collections of IFDCollection objects. This allows for a nesting of collections in meaningful ways. For example, the subclass IFDStructureDataAssociation (a subclass of IFDCollection) maintains two specific collections: one for structures, and one for data objects.

org.iupac.fairdata.core.IFDCollection inherits fields of IFDObject		
field name	data type	description
itemsByID	Map<String, IFDObject>	a Map of the items in this collection keyed to their id field
itemType	String	class name of all items (present only for where all items are the same type, allowing for individual items to not declare their itemType)
itemTypeExtends	String	a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to the first class name that is of the form <i>org.iupac.fairdata.core.*</i> (present only for where all items are the same type, allowing for individual items to not declare their itemType)

To be sure, an IFDCollection could represent a digital object in the form of a zip file (and usually does). Nonetheless, this does not make it "representable" in the sense that it is likely to have multiple distinctly different representations that characterize an IFDRepresentableObject. Instead, we would say that an IFDCollection is serializable, meaning it can be converted into string form.

2.3.6 IFDAssociation (abstract)

IFDAssociation is an abstract class that is a "collection of associated collections." This collection is intended to be "immutable" in the sense that some initial set of collections are associated, and then the collection is fixed. The number of items in the association is entirely customizable.

org.iupac.fairdata.core.IFDAssociation inherits fields of IFDCollection and IFDObject		
field name	data type	description

itemsByID	Map<String, String[]>	Each entry is a Map with keys derived from the id values of the contained associated collections (for example, "structures" and "spectra"), and with values that are arrays of id values of the items in those collections that are part of this association.
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2.3.7 IFDAssociationCollection (abstract)

IFDAssociationCollection objects are lists of IFDAssociation objects.

org.iupac.fairdata.core.IFDAssociationCollection inherits fields of IFDCollection and IFDObject		
field name	data type	description
collections	String[]	an array of the names of the finding aid's collections for which items are being associated, in the same order as the arrays in the IFDAssociation items
itemsByID	Map<String, IFDAssociation>	The associations in this collection, keyed by their id field

2.3.8 IFDRepresentableObject (abstract)

IFDRepresentableObject maintains a set of metadata relating to different ways a sample or structure or data object or analysis object might be represented. A class extending IFDRepresentableObject is expected to maintain a list of one or more distinctly different digital representations (byte sequences) that amount to more than just metadata.

org.iupac.fairdata.core.IFDRepresentableObject inherits fields of IFDObject		
field name	data type	description

representations	IFDRepresentation[]	an array of representations
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Note that there is no mechanism within the model for these classes to refer to each other. This is a key aspect of the IUPAC FAIRSpec Data Model. The model is based on the idea of a collection of independent objects, and it is the IFDAssociation class that accomplishes this referencing. This allows for a relatively simple and flexible packaging of objects, with their relationships identified only in key metadata resources, separately from the objects themselves.

2.3.9 IFDReference

IFDReference is the storage class connecting IFDRepresentations to their original source Digital Objects and their unique local name within an IUPAC FAIRData Collection as well as a DOI link to a landing page for the representation or a URL to the digital object itself (as in a repository, perhaps)

org.iupac.fairdata.core.IFDReference		
field name	data type	description
doi	String	the full DOI reference, including the protocol and host (for example, https://doi.org/10.14469/hpc/11741) generally to an HTML landing page describing this representation. (See also url .)
ifdType	String	the class name of this reference
ifdTypeExtends	String	a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to org.iupac.fairdata.core.IFDReference
localName	String	A file name to associate with this digital object; Typically a simple name such as “35.mol”; not present if localPath is present
localPath	String	the path to the digital object within the associated IFD FAIRSpec Data Collection; not present if localName is present
originPath	String	the path within the originating resource, if applicable
resourceID	String	the <i>id</i> of the IFDResource originating this reference
url	String	the URL string targeting this representation (for example, a repository file reference; see also doi .)

When *origin* is a ZIP file entry, *origin* will contain a "pipe" vertical line symbol "|" to indicate "a file within a zip file." Directories are represented with the standard forward slash. Thus, *origin* might be something like

1.zip|1_13C/pdata/1/thumb.png

indicating that the original digital item is an image several directories deep within the file *1.zip*. Note that this notation allows for resources that are from zip files that are themselves found within zip files.

Generally speaking, the *localName* will not be part of a zip file (other than perhaps the zip file used for the collection's delivery), and may even have no hierarchical directory structure, either. For example:

1.zip..1_13C_pdata_1_thumb.png

Note that an *IFDRepresentation* may or may not have an originating object, and its *IFDReference* may or may not have a *localPath* or an *originPath*. The lack of an *originPath* may indicate that an extractor has created this representation during the process of extraction as a "value-added" addition to the collection. For example, using just a SMILES string, the extractor may have created a 2D MOL file, a 3D MOL file, an InChI, and an InChIKey. The lack of a *localPath* indicates that a representation's byte data are being saved within the *IFDRepresentation* object itself, not separately, and serialized directly into the IUPAC FAIRData finding aid. In that case the *localName*, such as "Compound 36.mol", just proves a parser or viewing tool the option to offer a file name to the user or save the file under a meaningful file name.

2.3.10 IFDResource

IFDResource is a general-purpose structure for holding the byte length and reference for an external resource digital object. It is specifically used in the description of data sources from which an IUPAC FAIRData Collection is derived.

Fields of org.iupac.fairdata.core.IFDResource		
field name	data type	description
id	String	a unique identifier of this resource in the finding aid's list of resources
len	Integer	the byte length of this resource
ref	String	the URL of this resource, preferably a persistent identifier
ifdType	String	the class name of this resource, typically org.iupac.fairdata.core.IFDResource
ifdTypeExtends	String	a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to org.iupac.fairdata.core.IFDResource

2.3.11 IFDFindingAid and IFDCollectionSet

IFDFindingAid and **IFDCollectionSet** constitute the defining metadata objects for an IUPAC FAIRSpec Data Collection and its finding aid. The finding aid provides metadata relating to the entire collection. For example, details about a publication or thesis and source of data. In addition, it maintains one IFDCollectionSet. This collection itself is just a set of collections, consisting of a list of IFDRepresentableObject and IFDAssociation collections and associated metadata relating to the collection as a whole.

org.iupac.fairdata.core.IFDFindingAid inherits fields from IFDObject		
field name	source	data type
		description
relatedItems		Map<String, ?>[]
		An array of map elements that describe a subset of metadata items drawn from the metadata records of registered digital objects. For example, an array containing two maps, one containing metadata pertaining to a related publication (DOI, journal citation, publication title, author information) registered with CrossRef and one pertaining to the originating dataset (DOI, dataset title, contributor information) registered with DataCite.
collectionSet		IFDCollectionSet
		the set of collections in the IUPAC FAIRData collection described by this IUPAC FAIRData Finding Aid; this field is the serialized form of a single IFDCollectionSet
contents		Map<String, ?>
		a map outlining the contents of this finding aid, format specified by the implementor, primarily for human reference. <pre> ▼ contents: citationCount: 1 ▼ collections: ▼ 0: count: 30 name: "samples" type: "org.iupac.fairdata.sample.IFDSampleCollection" typeExtends: "org.iupac.fairdata.core.IFDCollection" ▶ 1: {...} ▶ 2: {...} ▶ 3: {...} resourceCount: 1 </pre>
created		Date
		a timestamp indicating when this finding aid was created, in the ISO-8601 UTC format (for example, 2019-05-09T17:15:35-07:00)

createdBy	String	an identifier of the software used to generate this finding aid
resources	IFDResource[]	a list of originating resources for this finding aid's collection
version	String	a semicolon-separated list of one or more relevant version identifiers

Note that the *contents* guide is for human readability and convenience only. Nothing in this section is essential to the finding aid itself or standardized for machine readability. Some implementations may provide only this section as the result of a preview query prior rather than replying with a fully elaborated finding aid.

org.iupac.fairdata.core.IFDCollectionSet inherits fields of IFDCollection and IFDObject		
field name	data type	description
itemsByID	Map<String, IFDCollection>	the top-level IFDCollections; for example with keys <i>samples, structures, spectra, compounds, analyses</i> .
resourceID	String	the <i>id</i> of the IFDResource given in IFDFindingAid.resources pointing to the IUPAC FAIRData Collection associated with this IFDFindingAid.

3.1 FAIRSpec Metadata object extensions

3.1.1 FAIRSpecFindingAid and FAIRSpecCollection classes

class FAIRSpecFindingAid extends IFDFindingAid

class FAIRSpecCollection extends IFDCollectionSet

FAIRSpecFindingAid adds a "brand" to an IFDFindingAid by declaring its type to be *org.iupac.fairdata.contrib.FAIRSpecFindingAid*. This indicates that this particular IFDFindingAid will contain metadata specific to the area of spectroscopy. When serialized, an object of this class becomes the IUPAC FAIRSpec Finding Aid digital object that ultimately distinguishes an IUPAC FAIRSpec Data Collection from other sorts of digital collections. The serialization of a FAIRSpecFindingAid object might be the target of a persistent identifier such as a DOI (Digital Object Identifier) for the collection (if XML) or interpreted by that DOI's landing page (if JSON). Or, it could be dynamically created in response to a query, allowing standardized programmatic access to the collection. A JSON serialization example is illustrated below and can be found at

<https://github.com/IUPAC/IUPAC-FAIRSpec/raw/main/examples/v4-preliminary/acs.joc.0c00770/IFD.findingaid.json>

In all cases, serialization of the finding aid must follow the guidelines of the context (either JSON or XML).

FAIRSpecCollection, like **FAIRSpecFindingAid**, brands the collection as utilizing spectroscopic-specific properties, reporting its type as *org.iupac.fairdata.contrib.FAIRSpecCollection*.

org.iupac.fairdata.contrib.fairspec.FAIRSpecFindingAid inherits fields from IFDFindingAid and IFDObject		
field name	data type	description
ifdType	String	"org.iupac.fairdata.contrib.fairspec.FAIRSpecFindingAid" or the class name of a subclass of this class.
ifdTypeExtends	String	"org.iupac.fairdata.core.IFDFindingAid" or a semicolon-separated list of superclass names of the ifdType terminating with <i>org.iupac.fairdata.contrib.fairspec.FAIRSpecFindingAid</i> ; <i>org.iupac.fairdata.core.IFDFindingAid</i>

org.iupac.fairdata.contrib.fairspec.FAIRSpecCollection inherits fields of IFDCollectionSet, IFDCollection, and IFDObject adds no additional fields

3.1.2 FAIRSpecDataObject and Instrumental technique-specific subclasses

class **FAIRSpecDataObject** extends IFDDataObject

The **FAIRSpecDataObject** is an abstract class that serves as the superclass identifying its subclasses as being spectroscopy related.

class **FAIRSpecCOMPData** extends **FAIRSpecDataObject**

class **FAIRSpecCOMPDataRepresentation** extends **IFDDataObjectRepresentation**

class **FAIRSpecIRData** extends **FAIRSpecDataObject**

class **FAIRSpecIRDataRepresentation** extends **IFDDataObjectRepresentation**

class **FAIRSpecNMRData** extends **FAIRSpecDataObject**

```
class FAIRSpecNMRDataRepresentation extends IFDDataObjectRepresentation
class FAIRSpecMSData extends FAIRSpecDataObject
class FAIRSpecMSDataRepresentation extends IFDDataObjectRepresentation
class FAIRSpecHRMSData extends FAIRSpecDataObject
class FAIRSpecHRMSDataRepresentation extends IFDDataObjectRepresentation
class FAIRSpecRamanData extends FAIRSpecDataObject
class FAIRSpecRamanDataRepresentation extends IFDDataObjectRepresentation
class FAIRSpecUVVISData extends FAIRSpecDataObject
class FAIRSpecUVVISDataRepresentation extends IFDDataObjectRepresentation
```

These technique-specific classes declare an IUPAC FAIRSpec object with additional technique-specific properties but do not add any new fields. Each subclass is associated with its related subclassed IFDDataObjectRepresentation (an actual spectrum, a PDF document, an image, etc.).

```
org.iupac.fairdata.contrib.fairspec.dataobject.ir.FAIRSpecCOMPData
org.iupac.fairdata.contrib.fairspec.dataobject.ir.FAIRSpecIRData
org.iupac.fairdata.contrib.fairspec.dataobject.nmr.FAIRSpecNMRData
org.iupac.fairdata.contrib.fairspec.dataobject.ms.FAIRSpecMSData
org.iupac.fairdata.contrib.fairspec.dataobject.hrms.FAIRSpecHRMSData
org.iupac.fairdata.contrib.fairspec.dataobject.raman.FAIRSpecRamanData
org.iupac.fairdata.contrib.fairspec.dataobject.uvvis.FAIRSpecUVVISData
```

inherits fields of IFDDataObject, and IFDObject
adds no additional fields

```
org.iupac.fairdata.contrib.fairspec.dataobject.comp.FAIRSpecCOMPDataRepresentation
org.iupac.fairdata.contrib.fairspec.dataobject.ir.FAIRSpecIRDataRepresentation
org.iupac.fairdata.contrib.fairspec.dataobject.nmr.FAIRSpecNMRDataRepresentation
org.iupac.fairdata.contrib.fairspec.dataobject.ms.FAIRSpecMSDataRepresentation
org.iupac.fairdata.contrib.fairspec.dataobject.hrms.FAIRSpecHRMSDataRepresentation
org.iupac.fairdata.contrib.fairspec.dataobject.raman.FAIRSpecRamanDataRepresentation
org.iupac.fairdata.contrib.fairspec.dataobject.uvvis.FAIRSpecUVVISDataRepresentation
```

inherits fields of IFDDataObjectRepresentation
adds no additional fields

3.1.3 FAIRSpecCompoundAssociation and FAIRSpecCompoundCollection classes

```
class FAIRSpecCompoundAssociation extends IFDStructureDataAssociation
class FAIRSpecCompoundCollection extends IFDStructureDataAssociationCollection
```

FAIRSpecCompoundAssociation represents an abstract association specifically involving one or more structures with zero or more spectra. The FAIRSpecCompoundAssociation is like all IFDAssociations in that it is a purely metadata object that has no direct representations as such. It

provides the ability to associate multiple structures (isomers of "Compound 3a", for example) with multiple spectra (1H NMR, 13C NMR, etc.). It allows for use of the term "compound" in its less restrictive common use in the literature as "one or more isomerically related compounds"). It could have a property that expresses this relationship, such as a `mInChI`, but this is not established in this version.

If there are no spectra, then the `FAIRSpecCompoundAssociation` becomes simply an association of isomerically related structures, much the same way a PubChem "compound" CID 21508 (2,3-dibromobutane) [<https://pubchem.ncbi.nlm.nih.gov/compound/2,3-dibromoButane>] is ambiguous, describing any one of four stereoisomers, or some mixture of them. However, the `FAIRSpecCompoundAssociation` contains an `IFDStructure` for each component, including one or more representations for each, individually. Additional context can be provided by the addition of properties or attributes that express the relationship and relative amounts of the related structures within the association.

In the case where the `FAIRSpecCompoundAssociation` includes one or more associated spectra, a single `IFDCompoundAssociation` can express a mixture of constitutionally unrelated chemical compounds that are associated with a set of spectra, "Compounds 3 and 4", for example, where 3 was the reactant and 4 was the product in a chemical reaction, and both are present in an NMR sample. In this case, the structures would have different properties or representations that unambiguously identify them as distinctly different chemical compounds. Future versions of the specification could add a property to `FAIRSpecCompoundAssociation` that would allow a quantitative description of the associated chemical compounds along the lines of a `mInChI` (mixture `InChI`).

The **`FAIRSpecCompoundCollection`** is a collection of `FAIRSpecCompoundAssociations` and forms a key element of an IUPAC FAIRSpec Data Collection, along with an `IFDStructureCollection` and an `IFDDataObjectCollection`.

`org.iupac.fairdata.contrib.fairspec.FAIRSpecCompoundAssociation`

inherits fields of `IFDStructureDataAssociation`, `IFDAssociation`, `IFDCollection`, and `IFDObject`
adds no additional fields

`org.iupac.fairdata.contrib.fairspec.FAIRSpecCompoundCollection`

inherits fields of `IFDStructureDataAssociationCollection`, `IFDCollection`, and `IFDObject`
adds no additional fields

3.1.4 Instrumental technique-specific properties

The IUPAC FAIRSpec extension adds a growing list of metadata key/value pairs specific to particular instrumental techniques. These definitions will evolve; current definitions can be found in the [IFD.properties file](#) at the IUPAC GitHub site. As of this version, for example, NMR properties include:

```
IFD.property.dataobject.fairspec.nmr.expt_dim
IFD.property.dataobject.fairspec.nmr.expt_offset_freq1
IFD.property.dataobject.fairspec.nmr.expt_offset_freq2
IFD.property.dataobject.fairspec.nmr.expt_offset_freq3
IFD.property.dataobject.fairspec.nmr.expt_nucl1
IFD.property.dataobject.fairspec.nmr.expt_nucl2
IFD.property.dataobject.fairspec.nmr.expt_nucl3
IFD.property.dataobject.fairspec.nmr.expt_pulse_prog
IFD.property.dataobject.fairspec.nmr.expt_solvent
IFD.property.dataobject.fairspec.nmr.expt_solvent_InChI
IFD.property.dataobject.fairspec.nmr.expt_solvent_InChIKey
IFD.property.dataobject.fairspec.nmr.expt_solvent_common_name
IFD.property.dataobject.fairspec.nmr.expt_thermodynamic_temperature
IFD.property.dataobject.fairspec.nmr.expt_title
IFD.property.dataobject.fairspec.nmr.instr_manufacturer_name
IFD.property.dataobject.fairspec.nmr.instr_probe_type
IFD.property.dataobject.fairspec.nmr.instr_proton_freq
IFD.property.dataobject.fairspec.nmr.instr_nominal_freq
```

Appendix A. Recognized IUPAC FAIRData Data Types

Data Type	Java Equivalent	Python Equivalent	C++ Equivalent
Byte	byte	byte	char
Integer	int or long*	int	int
Double	double	float	double
Boolean	boolean	bool	bool
String	String (UTF-8)	string (UTF-8)	std::string
byte	byte	byte	char
array	List or array*	list	array
Map	Map	dict	std::map

Date**	String	string	std:string
--------	--------	--------	------------

*The model does not make a distinction between specific implementations of data types such as int/long or List/array. In this specification, we will use them synonymously.

** Dates are Strings in the ISO-8601[<https://www.iso.org/iso-8601-date-and-time-format.html>] format (for example, 2019-05-09T17:15:35-07:00).

Appendix B. Core Classes

2.3.1 IFDObject (abstract)

IFDObject is an abstract superclass for nearly all IUPAC FAIRData Metadata Object Model classes.

- IFDObject
 - IFDCollection
 - IFDAssociation
 - IFDAssociationCollection
 - IFDCollectionSet
 - IFDSampleCollection
 - IFDStructureCollection
 - ...(several more)
 - IFDRepresentableObject
 - IFDAnalysisObject
 - IFDDataObject
 - IFDSample
 - IFDStructure
 - IFDFindingAid

Figure xxx. Subclasses of IFDObject in the IUPAC FAIRData Metadata Object Model.

org.iupac.fairdata.core.IFDOBJECT		
field name	data type	description
description	String	an optional
doi	String	the full DOI reference to this digital object, including the protocol and host (for example, https://doi.org/10.14469/hpc/11741); generally to an HTML landing page describing this object – for example, to a repository page for a collection. For an IFDRepresentableObject, each of the object's representations might have its own url. (See IFDReference .)

id	String	an optional unique identifier for this object; if present, must be unique within this finding aid
ifdType	String	required; the class name of this object, such as <i>org.iupac.fairdata.sample.IFDSample</i>
ifdTypeExtends	String	required for non-core classes; a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to the first class name that is of the form <i>org.iupac.fairdata.core.*</i> (only present for non-core classes)
label	String	an optional short descriptive but not necessarily unique label for this object
note	String	an optional free text note, possibly relating to issues in the production of the finding aid or collection; not intended for general metadata indexing
attributes	Map<String, String>	an optional set of key/value pairs that are not part of the IUPAC FAIRData standard or its extension. Characters of the keys must be one of [A-Za-z0-9_] or the space character and must be trimmed of any space character.
ifdProperties	Map<String, Object>	an optional set of key/value pairs where the key is a fully qualified IUPAC FAIRData property name starting with "IFD." or just its associated short (underscore-separated) name, if the propertyPrefix field is also present. The value must be of the form specified by this standard or its extension
propertyPrefix	String	optional; a prefix starting with "IFD." that should be prepended along with a "." to each property key not already starting with "IFD." in order to produce a fully qualified IUPAC FAIRData property name as specified by this standard or its extension
url	String	a URL string targeting this object's sole digital item. The URL field is only present if the doi field is empty and should not contain a DOI itself. For an IFDRepresentableObject, each of the object's representations might have its own url. (See IFDReference .)

IFDObject has three direct subclasses: IFDRepresentableObject, IFDCollection, and IFDFindingAid. An IFDRepresentableObject is a list of IFDRepresentation items, while an IFDCollection is a list of IFDRepresentableObject items. The IFDFindingAid is a special object that is not itself a collection but holds the IFDCollectionSet. These three subclasses are described in detail in section 2.3 *Core Classes*.

2.3.2 IFDRepresentation (abstract)

IFDRepresentation is an abstract class with instances that either reference actual digital objects or contain byte sequence or string data themselves. These may be images, spreadsheets, complex analyses, discipline-specific or vendor-specific file formats – any digital format that data (in the broad sense, including chemical structure "data") can be represented. All actual digital objects in an IUPAC FAIRData Collection will be represented in the model as an IFDRepresentation of one type or another and serialized as part of a List<IFDRepresentation>, where each representation in the list is in the form of a Map<String, String>. Note that IFDRepresentation objects themselves are not IFDObject objects and are not list-like. As an abstract class, IFDRepresentation can only be "realized" as a more specific sort of representation, for example:

- IFDRepresentation
 - IFDAnalysisObjectRepresentation
 - IFDDataObjectRepresentation
 - IFDSampleRepresentation
 - IFDStructureRepresentation

Figure xxx. Subclassing IFDRepresentation differentiates distinctly different types of representations.

Subclassing IFDRepresentation provides a way of ensuring that the correct general type of digital object in an IUPAC FAIRData Collection corresponds with a given IFDRepresentableObject. For example, chemical structure-related metadata objects (IFDStructure objects) must be lists of structure representations (IFDStructureRepresentation objects, such as MOL files). We say that IFDStructure (which is a List) extends IFDRepresentableObject<IFDStructureRepresentation>.

org.iupac.fairdata.core.IFDRepresentation		
field name	data type	description
data	String or byte[]	When the representation itself is suitably small in terms of bytes (in the opinion of the implementer), the representation can be serialized directly within the finding aid as a simple String when appropriate, for example, a SMILES string or InChI key. Alternatively, the value can take the form of a byte array and serialized as a base64-encoded String having the eight-character prefix ;base64,. Base64 encoding is described in RFC 4648. If the <i>data</i> field is not present or null, then the <i>ref</i> field must indicate an originPath or a localPath or a localName. If the <i>data</i> field is present and not null, then the <i>ref</i> field may or may not contain these references. For example, an InChI generated by a metadata extractor from a CDXML file will not have any local or origin path. An image stored within a

		Finding Aid may still have all of these.
ifdType	String	the object type of the form <i>IFD.representation...</i> declared in the extension of the specification
key	String	the name of this representation, starting with "IFD.representation." and continuing with a dot-separated set of qualifiers that consist only of lower-case or digit [a-z0-9] characters; for example: <i>IFD.representation.dataobject.fairspec.nmr.image</i>
len	Integer	the byte length of the digital object associated with this representation
mediaType	String	the media type [rfc....] of this digital object
ref	IFDReference	the machine-followable reference to the digital object associated with this reference, or <i>null</i> (without quotes) or absent when there is no such reference and the <i>data</i> field is non-null.

2.3.3 IFDProperty

IFDProperty is the storage class for standardized metadata key/value pairs for all IFDObjects. This class can be serialized in-line as a key:value pair within a Map<String,String> serialization.

org.iupac.fairdata.core.IFDProperty		
field name	data type	description
(key)	String	The name of this property, starting with "IFD.property." and continuing with a dot-separated set of qualifiers that consist only of lower-case or digit [a-z0-9] characters; for example: <i>IFD.property.dataobject.spec.nmr.expt.offset_freq1</i>
(value)	(variable)	the value of this metadata object, serialized as the data type and declared by the standard or its extension, in the units specified by the standard or its extension

Properties, in principle, may have specified data types (String, Integer, Integer, Float64, ...), units (Hz, MHz, Kelvin, ...), values, value ranges ([1...]), or value formats (1H, 13C, ...) that are specified in the standard or its extension. Properties may be serialized as key:value pairs within a map (as in JSON) or as attribute declarations such as *key=value* (as in XML).

2.3.3 IFDAttribute

IFDAttribute is the storage class for non-standardized metadata key/value pairs for all IFDObjects. An IFDAttributes can use one name to refer to a single value or a list of values.

org.iupac.fairdata.core.IFDProperty		
field name	data type	description
name	String	The name of this attribute consisting of a dot-separated sequence referring to a class followed by a dot and a sequence of lower-case or digit or underscore [a-z0-9] characters; for example: <i>IFD.property.collectionset.source_data_uri</i>
value	(variable)	the value of this metadata object, serialized as a number, boolean, string, or a list of such values.

2.3.5 IFDCollection

IFDCollection is an abstract class designed to hold a set of objects of the same class or superclass. IFDCollection objects are not themselves IFDRepresentableObjects and thus have no digital representation themselves. Note that IFDCollection objects may be collections of IFDCollection objects. This allows for a nesting of collections in meaningful ways. For example, the subclass IFDStructureDataAssociation (a subclass of IFDCollection) maintains two specific collections: one for structures, and one for data objects.

org.iupac.fairdata.core.IFDCollection inherits fields of IFDObject		
field name	data type	description
itemsByID	Map<String, IFDObject>	a Map of the items in this collection keyed to their id field
itemType	String	class name of all items (present only for where all items are the same type, allowing for individual items to not declare their itemType)
itemTypeExtends	String	a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to the first class name that is of the form <i>org.iupac.fairdata.core.*</i> (present only for where all items are the same type, allowing for individual items to not declare their itemType)

To be sure, an IFDCollection could represent a digital object in the form of a zip file (and usually does). Nonetheless, this does not make it "representable" in the sense that it is likely to have multiple distinctly different representations that characterize an IFDRepresentableObject. Instead, we would say that an IFDCollection is serializable, meaning it can be converted into string form.

2.3.6 IFDAssociation (abstract)

IFDAssociation is an abstract class that is a "collection of associated collections." The collection is intended to be "immutable" in the sense that some initial set of collections are associated, and then the collection is fixed. The number of items in the association is entirely customizable.

org.iupac.fairdata.core.IFDAssociation inherits fields of IFDCollection and IFDObject		
field name	data type	description

itemsByID	Map<String, String[]>	Each entry is a Map with keys derived from the id values of the contained associated collections (for example, "structures" and "spectra"), and with values that are arrays of id values of the items in those collections that are part of this association.
-----------	------------------------	---

IFDAssociation is particularly used for metadata that relates one object to another – a sample to a set of spectra; a set of spectra to a structure. An IFDAssociation in its simplest serialized form is just a set of pointers to representable objects in other collections of the FAIRData Finding Aid.

IFDAssociations are serialized as the items of an IFDAssociationCollection. Thus, in serialized form, an association of samples and their spectra might appear as shown below:

```
"compounds":{
  "id":"compounds",
  "itemType":"org.iupac.fairdata.contrib.fairspec.FAIRSpecCompound",
  "itemTypeExtends":"org.iupac.fairdata.derived.IFDStructureDataAssociation;org.iupac.fairdata.core.IFDAssociation",
  "itemsByID": {
    "3aa" : {
      "id":"3aa",
      "itemsByID": {
        "structures":["3aa"],
        "spectra":["3aa-C","3aa-H"]
      }
    },
    ...
  }
}
```

Thus, associations essentially provide the "edges" to a graph having IFDRepresentableObject nodes. The byID option is preferred, allowing for a much more human-readable finding aid.

2.3.7 IFDAssociationCollection (abstract)

IFDAssociationCollection objects are lists of IFDAssociation objects.

org.iupac.fairdata.core.IFDAssociationCollection inherits fields of IFDCollection and IFDObject		
field name	data type	description

collections	String[]	an array of the names of the finding aid's collections for which items are being associated, in the same order as the arrays in the IFDAssociation items
itemsByID	Map<String, FIDAssociation>	The associations in this collection, keyed by their id field

2.3.8 IFDRepresentableObject (abstract)

IFDRepresentableObject maintains a set of metadata relating to different ways a sample or structure or data object or analysis object might be represented. A class extending IFDRepresentableObject is expected to maintain a list of one or more distinctly different digital representations (byte sequences) that amount to more than just metadata.

org.iupac.fairdata.core.IFDRepresentableObject inherits fields of IFDObject		
field name	data type	description
representations	IFDRepresentation[]	an array of representations

Note that there is no mechanism within the model for these classes to refer to each other. This is a key aspect of the IUPAC FAIRSpec Data Model. The model is based on the idea of a collection of independent objects, and it is the IFDAssociation class that accomplishes this referencing. This allows for a relatively simple and flexible packaging of objects, with their relationships identified only in key metadata resources, separately from the objects themselves.

2.3.9 IFDReference

IFDReference is the storage class connecting IFDRepresentations to their original source Digital Objects and their unique local name within an IUPAC FAIRData Collection as well as a DOI link to a landing page for the representation or a URL to the digital object itself (as in a repository, perhaps)

org.iupac.fairdata.core.IFDReference		
field name	data type	description
doi	String	the full DOI reference, including the protocol and host (for example, https://doi.org/10.14469/hpc/11741) generally to an HTML landing page describing this representation.

		(See also url .)
ifdType	String	the class name of this reference
ifdTypeExtends	String	a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to org.iupac.fairdata.core.IFDReference
localName	String	A file name to associate with this digital object; Typically a simple name such as "35.mol"; not present if localPath is present
localPath	String	the path to the digital object within the associated IUPAC FAIRSpec Data Collection; not present if localName is present
originPath	String	the path within the originating resource, if applicable
resourceID	String	the <i>id</i> of the IFDResource originating this reference
url	String	the URL string targeting this representation (for example, a repository file reference; see also doi .)

When *origin* is a ZIP file entry, *origin* will contain a "pipe" vertical line symbol "|" to indicate "a file within a zip file." Directories are represented with the standard forward slash. Thus, *origin* might be something like

1.zip|1_13C/pdata/1/thumb.png

indicating that the original digital item is an image several directories deep within the file *1.zip*. Note that this notation allows for resources that are from zip files that are themselves found within zip files.

Generally speaking, the localName will not be part of a zip file (other than perhaps the zip file used for the collection's delivery), and may even have no hierarchical directory structure, either. For example:

1.zip..1_13C_pdata_1_thumb.png

Note that an IFDRepresentation may or may not have an originating object, and its IFDReference may or may not have a localPath or an originPath. The lack of an originPath may indicate that an extractor has created this representation during the process of extraction as a "value-added" addition to the collection. For example, using just a SMILES string, the extractor may have created a 2D MOL file, a 3D MOL file, an InChI, and an InChIKey. The lack of a localPath indicates that a representation's byte data are being saved within the IFDRepresentation object itself, not separately, and serialized directly into the IUPAC FAIRData finding aid. In that case the localName, such as "Compound 36.mol", just proves a parser or viewing tool the option to offer a file name to the user or save the file under a meaningful file name.

2.3.10 IFDResource

IFDResource is a general-purpose structure for holding the byte length and reference for an external resource digital object. It is specifically used in the description of data sources from which an IUPAC FAIRData Collection is derived.

Fields of org.iupac.fairdata.core.IFDResource		
field name	data type	description
id	String	a unique identifier of this resource in the finding aid's list of resources
len	Integer	the byte length of this resource
ref	String	the URL of this resource, preferably a persistent identifier
ifdType	String	the class name of this resource, typically org.iupac.fairdata.core.IFDResource
ifdTypeExtends	String	a semicolon-separated list of names of classes in the superclass hierarchy of this class, up to org.iupac.fairdata.core.IFDResource

2.3.11 IFDFindingAid and IFDCollectionSet

IFDFindingAid and **IFDCollectionSet** constitute the defining metadata objects for an IUPAC FAIRSpec Data Collection and its finding aid. The finding aid provides metadata relating to the entire collection. For example, details about a publication or thesis and source of data. In addition, it maintains one IFDCollectionSet. This collection itself is just a set of collections, consisting of a list of IFDRepresentableObject and IFDAssociation collections and associated metadata relating to the collection as a whole.

org.iupac.fairdata.core.IFDFindingAid inherits fields from IFDObject		
field name	data type	description
relatedItems	Map<String,?>[]	An array of map elements that describe a subset of metadata items drawn from the metadata records of registered digital objects. For example, an array containing two maps, one containing metadata pertaining to a related publication (DOI, journal citation, publication title, author information)

		registered with CrossRef and one pertaining to the originating dataset (DOI, dataset title, contributor information) registered with DataCite.
collectionSet	IFDCollectionSet	the set of collections in the IUPAC FAIRData collection described by this IUPAC FAIRData Finding Aid; this field is the serialized form of a single IFDCollectionSet
contents	Map<String, ?>	a map outlining the contents of this finding aid, format specified by the implementor, primarily for human reference. <pre> ▼ contents: citationCount: 1 ▼ collections: ▼ 0: count: 30 name: "samples" type: "org.iupac.fairdata.sample.IFDSampleCollection" typeExtends: "org.iupac.fairdata.core.IFDCollection" ▶ 1: {...} ▶ 2: {...} ▶ 3: {...} resourceCount: 1 </pre>
created	Date	a timestamp indicating when this finding aid was created, in the ISO-8601 UTC format (for example, 2019-05-09T17:15:35-07:00)
createdBy	String	an identifier of the software used to generate this finding aid
resources	IFDResource[]	a list of originating resources for this finding aid's collection
version	String	a semicolon-separated list of one or more relevant version identifiers

Note that the *contents* guide is for human readability and convenience only. Nothing in this section is essential to the finding aid itself or standardized for machine readability. Some implementations may provide only this section as the result of a preview query prior rather than replying with a fully elaborated finding aid.

org.iupac.fairdata.core.IFDCollectionSet inherits fields of IFDCollection and IFDObject		
field name	data type	description
itemsByID	Map<String, IFDCollection>	the top-level IFDCollections; for example with keys <i>samples</i> , <i>structures</i> , <i>spectra</i> , <i>compounds</i> , <i>analyses</i> .

resourceID	String	the <i>id</i> of the IFDResource given in IFDFindingAid.resources pointing to the IUPAC FAIRData Collection associated with this IFDFindingAid.
------------	--------	---

2.5 Derived subclasses

These classes listed in this section supply the connecting links among samples, data objects, chemical structures, and post-acquisition data analysis. Whereas an association can have any (fixed) number of elements (in the form of IFDCollection objects), the associations presented here involve only two or three elements. Only representative examples of these classes are described below.

2.5.1 two-element IFDSample-IFDStructure-IFDDataObject associations

class IFDSampleDataAssociation extends IFDAssociation

class IFDSampleDataAssociationCollection extends IFDAssociationCollection

class IFDStructureDataAssociation extends IFDAssociation

class IFDStructureDataAssociationCollection extends IFDAssociationCollection

class IFDSampleStructureAssociation extends IFDAssociation

class IFDSampleStructureAssociationCollection extends IFDAssociationCollection

2.5.1.1 IFDSampleDataAssociation

This class links an experimental sample with its associated data without any explicit association with a chemical structure. It contains two collections, an IFDSampleCollection and an IFDDataObjectCollection. (Typically the sample collection would consist of only one IFDSample object, but we allow for the possibility that, for example, multiple samples have contributed to the linked data object collection.

2.5.1.2 IFDStructureDataAssociation

This class is the key class for connecting a chemical structure with its spectroscopic data, containing two collections, an IFDStructureCollection and an IFDDataObjectCollection. In the simplest case, this would be one structure and its associated spectrum, but it can represent all possible relationships: one-to-one (the structure associated with a spectrum), one-to-many (a set of spectra associated with a give structure), many-to-one (a set of structures describing a mixture associated with a spectrum), and many-to-many (a set of spectra associated with a mixture of compounds).

2.5.1.3 IFDSampleStructureAssociation

This class indicates the connection between (one or more) physical samples and (one or more) chemical structures. It is intended to be used specifically when there is no data object in the collection that links a given sample with a given structure, or when there is no such data *needed* to make that association. For example, the sample is a commercial sample, and the quality of that sample can be ensured. Or the sample's chemical identity is known, but the linking data are not available. or the way the digital entities were ingested into the collection made the logical connection between a "sample" (perhaps the "compound number" of a journal article") and one or more chemical structures (drawn by the author and given that specific compound number).

2.5.2 three-element "analysis" associations

class IFDStructureDataAnalysis extends IFDAssociation

class IFDStructureDataAnalysisCollection extends IFDAssociationCollection

2.5.2.1 IFDStructureDataAnalysis

org.iupac.fairdata.derived.IFDSampleDataAssociation
org.iupac.fairdata.derived.IFDSampleStructureAssociation
org.iupac.fairdata.derived.IFDStructureDataAssociation
org.iupac.fairdata.derived.IFDStructureDataAnalysis

inherits fields of IFDAssociation and IFDObject
adds no additional fields

org.iupac.fairdata.derived.IFDSampleDataAssociationCollection
org.iupac.fairdata.derived.IFDSampleStructureAssociationCollection
org.iupac.fairdata.derived.IFDStructureDataAssociationCollection
org.iupac.fairdata.derived.IFDStructureDataAnalysisCollection

inherits fields of IFDAssociationCollection, IFDCollection, and IFDObject
adds no additional fields

Appendix C. Specialized subclasses

Appendix D. Full list of IUPAC FAIRData MetaData-Related Classes

B.1 core classes (11)

org.iupac.fairdata.core.IFDAssociation
org.iupac.fairdata.core.IFDAssociationCollection
org.iupac.fairdata.core.IFDCollection
org.iupac.fairdata.core.IFDCollectionSet
org.iupac.fairdata.core.IFDFindingAid
org.iupac.fairdata.core.IFDOBJECT
org.iupac.fairdata.core.IFDProperty
org.iupac.fairdata.core.IFDReference
org.iupac.fairdata.core.IFDRepresentableObject
org.iupac.fairdata.core.IFDRepresentation
org.iupac.fairdata.core.IFDResource

B.2 primary subclasses of IFDRepresentableObject, and related classes (12)

org.iupac.fairdata.analysis.IFDAnalysisObject
org.iupac.fairdata.analysis.IFDAnalysisObjectCollection
org.iupac.fairdata.analysis.IFDAnalysisObjectRepresentation

org.iupac.fairdata.dataobject.IFDDataObject
org.iupac.fairdata.dataobject.IFDDataObjectCollection
org.iupac.fairdata.dataobject.IFDDataObjectRepresentation

org.iupac.fairdata.sample.IFDSample
org.iupac.fairdata.sample.IFDSampleCollection
org.iupac.fairdata.sample.IFDSampleRepresentation

org.iupac.fairdata.structure.IFDStructure
org.iupac.fairdata.structure.IFDStructureCollection
org.iupac.fairdata.structure.IFDStructureRepresentation

B.3 derived associations and related classes (8)

org.iupac.fairdata.derived.IFDSampleDataAssociation
org.iupac.fairdata.derived.IFDSampleDataAssociationCollection

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org.iupac.fairdata.derived.IFDSampleStructureAssociation
org.iupac.fairdata.derived.IFDSampleStructureAssociationCollection

org.iupac.fairdata.derived.IFDStructureDataAnalysis
org.iupac.fairdata.derived.IFDStructureDataAnalysisCollection

org.iupac.fairdata.derived.IFDStructureDataAssociation
org.iupac.fairdata.derived.IFDStructureDataAssociationCollection

Appendix C. Current list of IUPAC FAIRData Property and Representation Metadata Keys

This list is derived from the two Java property files, [ifd.properties](#) and [fairspec.properties](#). They take the form property definitions:

<property_name>=<metadata key>

These definitions can be used by an extractor to generate an IUPAC FAIRSpec Finding Aid.

collection set and structure properties

IFD_PROPERTY_COLLECTIONSET_BYID=IFD.property.collectionset.byid
IFD_PROPERTY_COLLECTIONSET_SOURCE_REPOSITORY_URI=IFD.property.collectionset.source_repository_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_PUBLICATION_URI=IFD.property.collectionset.source_publication_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_DATA_URI=IFD.property.collectionset.source_data_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_DATA_LICENSE_URI=IFD.property.collectionset.source_data_license_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_DATA_LICENSE_NAME=IFD.property.collectionset.source_data_license_name
IFD_PROPERTY_COLLECTIONSET_REF=IFD.property.collectionset.ref
IFD_PROPERTY_COLLECTIONSET_LEN=IFD.property.collectionset.len

IFD_PROPERTY_STRUCTURE_INCHIKEY=IFD.property.structure.inchikey
IFD_PROPERTY_STRUCTURE_MOLECULAR_FORMULA=IFD.property.structure.molecular_formula
IFD_PROPERTY_STRUCTURE_EMPIRICAL_FORMULA=IFD.property.structure.empirical_formula
IFD_PROPERTY_STRUCTURE_CELL_FORMULA=IFD.property.structure.cell_formula

IFD_PROPERTY_DATAOBJECT_ORIGINATING_SAMPLE_ID=IFD.property.dataobject.originating_sample_id
IFD_PROPERTY_DATAOBJECT_TIMESTAMP=IFD.property.dataobject.timestamp

core structure representations

IFD_REP_STRUCTURE_MOL=IFD.representation.structure.mol
IFD_REP_STRUCTURE_SDF=IFD.representation.structure.sdf
IFD_REP_STRUCTURE_CDX=IFD.representation.structure.cdx
IFD_REP_STRUCTURE_CDXML=IFD.representation.structure.cdxml
IFD_REP_STRUCTURE_CML=IFD.representation.structure.cml

IFD_REP_STRUCTURE_MOL_2D=IFD.representation.structure.mol_2d
IFD_REP_STRUCTURE_MOL_3D=IFD.representation.structure.mol_3d
IFD_REP_STRUCTURE_SDF_2D=IFD.representation.structure.sdf_2d
IFD_REP_STRUCTURE_SDF_3D=IFD.representation.structure.sdf_3d

IFD_REP_STRUCTURE_PNG=IFD.representation.structure.png
IFD_REP_STRUCTURE_UNKNOWN=IFD.representation.structure.unknown
IFD_REP_STRUCTURE_SMILES=IFD.representation.structure.smiles
IFD_REP_STRUCTURE_INCHI=IFD.representation.structure.inchi
IFD_REP_STRUCTURE_STANDARD_INCHI=IFD.representation.structure.standard_inchi
IFD_REP_STRUCTURE_FIXEDH_INCHI=IFD.representation.structure.fixedh_inchi
IFD_REP_STRUCTURE_CIF=IFD.representation.structure.cif

current and proposed media types

IFD_MEDIATYPE_CDX=chemical/x-cdx
IFD_MEDIATYPE_CDXML=chemical/x-cdxml
IFD_MEDIATYPE_CIF=chemical/x-cif
IFD_MEDIATYPE_INCHI=chemical/x-inchi
IFD_MEDIATYPE_JDX=chemical/x-jcamp-dx
IFD_MEDIATYPE_JPF=application/octet-stream;chemical/x-nmr-jeol # (proposed)
IFD_MEDIATYPE_MNOVA=application/octet-stream;chemical/x-mnova # (proposed)

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IFD_MEDIATYPE_MOL=chemical/x-mdl-molfile
IFD_MEDIATYPE_SDF=chemical/x-mdl-sdfile
IFD_MEDIATYPE_SMILES=chemical/x-daylight-smiles
IFD_MEDIATYPE_SMI=chemical/x-daylight-smiles
IFD_MEDIATYPE_CML=chemical/x-cml

IFD_MEDIATYPE_GJF=chemical/x-gjf # (proposed) Gaussian input file
IFD_MEDIATYPE_FCHK=chemical/x-fchk # (proposed) Gaussian check file

IFD_MEDIATYPE_JPG=image/jpeg
IFD_MEDIATYPE_PNG=image/png
IFD_MEDIATYPE_LOG=text/plain
IFD_MEDIATYPE_TXT=text/plain
IFD_MEDIATYPE_OUT=text/plain
IFD_MEDIATYPE_PDF=application/pdf
IFD_MEDIATYPE_ZIP=application/zip

fairspec properties

actionable IFDDataObject types

DATAOBJECT_FAIRSPEC_HRMS_FLAG=.dataobject.fairspec.hrms.
DATAOBJECT_FAIRSPEC_IR_FLAG=.dataobject.fairspec.ir.
DATAOBJECT_FAIRSPEC_NMR_FLAG=.dataobject.fairspec.nmr.
DATAOBJECT_FAIRSPEC_RAMAN_FLAG=.dataobject.fairspec.raman.
DATAOBJECT_FAIRSPEC_UVVIS_FLAG=.dataobject.fairspec.uvvis.
DATAOBJECT_FAIRSPEC_COMP_FLAG=.dataobject.fairspec.comp.
DATAOBJECT_FAIRSPEC_XRAY_FLAG=.dataobject.fairspec.xray.
DATAOBJECT_FAIRSPEC_UNKNOWN_FLAG=.dataobject.fairspec.unknown.

properties

IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_THERMODYNAMIC_TEMPERATURE=IFD.property.dataobject.fairspec.nmr.expt_thermodynamic_temperature
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_DIMENSION=IFD.property.dataobject.fairspec.nmr.expt_dimension
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_FREQ_1=IFD.property.dataobject.fairspec.nmr.expt_offset_freq1
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_FREQ_2=IFD.property.dataobject.fairspec.nmr.expt_offset_freq2
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_FREQ_3=IFD.property.dataobject.fairspec.nmr.expt_offset_freq3
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_NUCL_1=IFD.property.dataobject.fairspec.nmr.expt_nucl1
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_NUCL_2=IFD.property.dataobject.fairspec.nmr.expt_nucl2
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_NUCL_3=IFD.property.dataobject.fairspec.nmr.expt_nucl3
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_PULSE_PROG=IFD.property.dataobject.fairspec.nmr.expt_pulse_prog
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT=IFD.property.dataobject.fairspec.nmr.expt_solvent
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT_InChI=IFD.property.dataobject.fairspec.nmr.expt_solvent_InChI
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT_InChIKey=IFD.property.dataobject.fairspec.nmr.expt_solvent_InChIKey
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT_COMMON_NAME=IFD.property.dataobject.fairspec.nmr.expt_solvent_common_name
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_TITLE=IFD.property.dataobject.fairspec.nmr.expt_title
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_PROC_TIMESTAMP=IFD.property.dataobject.fairspec.nmr.proc_timestamp
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_NOMINAL_FREQ=IFD.property.dataobject.fairspec.nmr.instr_proton_freq
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_NOMINAL_FREQ=IFD.property.dataobject.fairspec.nmr.instr_nominal_freq
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_MANUFACTURER_NAME=IFD.property.dataobject.fairspec.nmr.instr_manufacturer_name
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_PROBE_TYPE=IFD.property.dataobject.fairspec.nmr.instr_probe_type

[# representations](#)

IFD_REP_DATAOBJECT_FAIRSPEC_HRMS_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.hrms.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_HRMS_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.hrms.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_HRMS_VENDOR_DATASET=IFD.representation.dataobject.fairspec.hrms.vendor_dataset

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IFD_REP_DATAOBJECT_FAIRSPEC_IR_JCAMP=IFD.representation.dataobject.fairspec.ir.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.ir.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.ir.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.ir.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_TEXT=IFD.representation.dataobject.fairspec.ir.spectrum_text
IFD_REP_DATAOBJECT_FAIRSPEC_IR_VENDOR_DATASET=IFD.representation.dataobject.fairspec.ir.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_MS_JCAMP=IFD.representation.dataobject.fairspec.ms.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.ms.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.ms.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.ms.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_TEXT=IFD.representation.dataobject.fairspec.ms.spectrum_text
IFD_REP_DATAOBJECT_FAIRSPEC_MS_VENDOR_DATASET=IFD.representation.dataobject.fairspec.ms.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_FID_1D=IFD.representation.dataobject.fairspec.nmr.jcamp_fid_1d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_FID_2D=IFD.representation.dataobject.fairspec.nmr.jcamp_fid_2d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_1i1r_1D=IFD.representation.dataobject.fairspec.nmr.jcamp_1i1r_1d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_1r_1D=IFD.representation.dataobject.fairspec.nmr.jcamp_1r_1d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_2D=IFD.representation.dataobject.fairspec.nmr.jcamp_2d

IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.nmr.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_TEXT=IFD.representation.dataobject.fairspec.nmr.spectrum_text
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.nmr.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.nmr.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_VENDOR_DATASET=IFD.representation.dataobject.fairspec.nmr.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_JCAMP=IFD.representation.dataobject.fairspec.raman.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.raman.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.raman.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.raman.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_VENDOR_DATASET=IFD.representation.dataobject.fairspec.raman.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_JCAMP=IFD.representation.dataobject.fairspec.uvvis.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.uvvis.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_DESCRIPTION=IFD.representation.dataobject.fairspec.uvvis.spectrum_description
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.uvvis.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.uvvis.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_VENDOR_DATASET=IFD.representation.dataobject.fairspec.uvvis.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_COMP_INPUT=IFD.representation.dataobject.fairspec.comp.input
IFD_REP_DATAOBJECT_FAIRSPEC_COMP_OUTPUT=IFD.representation.dataobject.fairspec.comp.output

IFD_REP_DATAOBJECT_FAIRSPEC_XRAY_CIF=IFD.representation.dataobject.fairspec.xray.cif

Appendix D. Full list of IUPAC FAIRData MetaData-Related Classes

B.1 core classes (11)

org.iupac.fairdata.core.IFDAssociation
org.iupac.fairdata.core.IFDAssociationCollection
org.iupac.fairdata.core.IFDCollection
org.iupac.fairdata.core.IFDCollectionSet

org.iupac.fairdata.core.IFDFindingAid
org.iupac.fairdata.core.IFDObject
org.iupac.fairdata.core.IFDProperty
org.iupac.fairdata.core.IFDReference
org.iupac.fairdata.core.IFDRepresentableObject
org.iupac.fairdata.core.IFDRepresentation
org.iupac.fairdata.core.IFDResource

B.2 primary subclasses of IFDRepresentableObject, and related classes (12)

org.iupac.fairdata.analysis.IFDAnalysisObject
org.iupac.fairdata.analysis.IFDAnalysisObjectCollection
org.iupac.fairdata.analysis.IFDAnalysisObjectRepresentation

org.iupac.fairdata.dataobject.IFDDataObject
org.iupac.fairdata.dataobject.IFDDataObjectCollection
org.iupac.fairdata.dataobject.IFDDataObjectRepresentation

org.iupac.fairdata.sample.IFDSample
org.iupac.fairdata.sample.IFDSampleCollection
org.iupac.fairdata.sample.IFDSampleRepresentation

org.iupac.fairdata.structure.IFDStructure
org.iupac.fairdata.structure.IFDStructureCollection
org.iupac.fairdata.structure.IFDStructureRepresentation

B.3 derived associations and related classes (8)

org.iupac.fairdata.derived.IFDSampleDataAssociation
org.iupac.fairdata.derived.IFDSampleDataAssociationCollection

org.iupac.fairdata.derived.IFDSampleStructureAssociation
org.iupac.fairdata.derived.IFDSampleStructureAssociationCollection

org.iupac.fairdata.derived.IFDStructureDataAnalysis
org.iupac.fairdata.derived.IFDStructureDataAnalysisCollection

org.iupac.fairdata.derived.IFDStructureDataAssociation
org.iupac.fairdata.derived.IFDStructureDataAssociationCollection

Appendix C. Current list of IUPAC FAIRData Property and Representation Metadata Keys

This list is derived from the two Java property files, [ifd.properties](#) and [fairspec.properties](#). They take the form property definitions:

<property_name>=<metadata key>

These definitions can be used by an extractor to generate an IUPAC FAIRSpec Finding Aid.

collection set and structure properties

IFD_PROPERTY_COLLECTIONSET_BYID=IFD.property.collectionset.byid
IFD_PROPERTY_COLLECTIONSET_SOURCE_REPOSITORY_URI=IFD.property.collectionset.source_repository_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_PUBLICATION_URI=IFD.property.collectionset.source_publication_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_DATA_URI=IFD.property.collectionset.source_data_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_DATA_LICENSE_URI=IFD.property.collectionset.source_data_license_uri
IFD_PROPERTY_COLLECTIONSET_SOURCE_DATA_LICENSE_NAME=IFD.property.collectionset.source_data_license_name
IFD_PROPERTY_COLLECTIONSET_REF=IFD.property.collectionset.ref
IFD_PROPERTY_COLLECTIONSET_LEN=IFD.property.collectionset.len

IFD_PROPERTY_STRUCTURE_INCHIKEY=IFD.property.structure.inchikey
IFD_PROPERTY_STRUCTURE_MOLECULAR_FORMULA=IFD.property.structure.molecular_formula
IFD_PROPERTY_STRUCTURE_EMPIRICAL_FORMULA=IFD.property.structure.empirical_formula
IFD_PROPERTY_STRUCTURE_CELL_FORMULA=IFD.property.structure.cell_formula

IFD_PROPERTY_DATAOBJECT_ORIGINATING_SAMPLE_ID=IFD.property.dataobject.originating_sample_id
IFD_PROPERTY_DATAOBJECT_TIMESTAMP=IFD.property.dataobject.timestamp

core structure representations

IFD_REP_STRUCTURE_MOL=IFD.representation.structure.mol
IFD_REP_STRUCTURE_SDF=IFD.representation.structure.sdf
IFD_REP_STRUCTURE_CDX=IFD.representation.structure.cdx
IFD_REP_STRUCTURE_CDXML=IFD.representation.structure.cdxml
IFD_REP_STRUCTURE_CML=IFD.representation.structure.cml

IFD_REP_STRUCTURE_MOL_2D=IFD.representation.structure.mol_2d
IFD_REP_STRUCTURE_MOL_3D=IFD.representation.structure.mol_3d
IFD_REP_STRUCTURE_SDF_2D=IFD.representation.structure.sdf_2d
IFD_REP_STRUCTURE_SDF_3D=IFD.representation.structure.sdf_3d

IFD_REP_STRUCTURE_PNG=IFD.representation.structure.png
IFD_REP_STRUCTURE_UNKNOWN=IFD.representation.structure.unknown
IFD_REP_STRUCTURE_SMILES=IFD.representation.structure.smiles
IFD_REP_STRUCTURE_INCHI=IFD.representation.structure.inchi
IFD_REP_STRUCTURE_STANDARD_INCHI=IFD.representation.structure.standard_inchi
IFD_REP_STRUCTURE_FIXEDH_INCHI=IFD.representation.structure.fixedh_inchi
IFD_REP_STRUCTURE_CIF=IFD.representation.structure.cif

current and proposed media types

IFD_MEDIATYPE_CDX=chemical/x-cdx
IFD_MEDIATYPE_CDXML=chemical/x-cdxml
IFD_MEDIATYPE_CIF=chemical/x-cif
IFD_MEDIATYPE_INCHI=chemical/x-inchi
IFD_MEDIATYPE_JDX=chemical/x-jcamp-dx
IFD_MEDIATYPE_JPF=application/octet-stream;chemical/x-nmr-jeol # (proposed)
IFD_MEDIATYPE_MNOVA=application/octet-stream;chemical/x-mnova # (proposed)

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IFD_MEDIATYPE_MOL=chemical/x-mdl-molfile
IFD_MEDIATYPE_SDF=chemical/x-mdl-sdfile
IFD_MEDIATYPE_SMILES=chemical/x-daylight-smiles
IFD_MEDIATYPE_SMI=chemical/x-daylight-smiles
IFD_MEDIATYPE_CML=chemical/x-cml

IFD_MEDIATYPE_GJF=chemical/x-gjf # (proposed) Gaussian input file
IFD_MEDIATYPE_FCHK=chemical/x-fchk # (proposed) Gaussian check file

IFD_MEDIATYPE_JPG=image/jpeg
IFD_MEDIATYPE_PNG=image/png
IFD_MEDIATYPE_LOG=text/plain
IFD_MEDIATYPE_TXT=text/plain
IFD_MEDIATYPE_OUT=text/plain
IFD_MEDIATYPE_PDF=application/pdf
IFD_MEDIATYPE_ZIP=application/zip

fairspec properties

actionable IFDDataObject types

DATAOBJECT_FAIRSPEC_HRMS_FLAG=.dataobject.fairspec.hrms.
DATAOBJECT_FAIRSPEC_IR_FLAG=.dataobject.fairspec.ir.
DATAOBJECT_FAIRSPEC_NMR_FLAG=.dataobject.fairspec.nmr.
DATAOBJECT_FAIRSPEC_RAMAN_FLAG=.dataobject.fairspec.raman.
DATAOBJECT_FAIRSPEC_UVVIS_FLAG=.dataobject.fairspec.uvvis.
DATAOBJECT_FAIRSPEC_COMP_FLAG=.dataobject.fairspec.comp.
DATAOBJECT_FAIRSPEC_XRAY_FLAG=.dataobject.fairspec.xray.
DATAOBJECT_FAIRSPEC_UNKNOWN_FLAG=.dataobject.fairspec.unknown.

properties

IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_THERMODYNAMIC_TEMPERATURE=IFD.property.dataobject.fairspec.nmr.expt_thermodynamic_temperature
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_DIMENSION=IFD.property.dataobject.fairspec.nmr.expt_dimension
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_FREQ_1=IFD.property.dataobject.fairspec.nmr.expt_offset_freq1
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_FREQ_2=IFD.property.dataobject.fairspec.nmr.expt_offset_freq2
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_FREQ_3=IFD.property.dataobject.fairspec.nmr.expt_offset_freq3
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_NUCL_1=IFD.property.dataobject.fairspec.nmr.expt_nucl1
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_NUCL_2=IFD.property.dataobject.fairspec.nmr.expt_nucl2
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_NUCL_3=IFD.property.dataobject.fairspec.nmr.expt_nucl3
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_PULSE_PROG=IFD.property.dataobject.fairspec.nmr.expt_pulse_prog
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT=IFD.property.dataobject.fairspec.nmr.expt_solvent
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT_InChI=IFD.property.dataobject.fairspec.nmr.expt_solvent_InChI
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT_InChIKey=IFD.property.dataobject.fairspec.nmr.expt_solvent_InChIKey
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_SOLVENT_COMMON_NAME=IFD.property.dataobject.fairspec.nmr.expt_solvent_common_name
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_EXPT_TITLE=IFD.property.dataobject.fairspec.nmr.expt_title
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_PROC_TIMESTAMP=IFD.property.dataobject.fairspec.nmr.proc_timestamp
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_NOMINAL_FREQ=IFD.property.dataobject.fairspec.nmr.instr_proton_freq
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_NOMINAL_FREQ=IFD.property.dataobject.fairspec.nmr.instr_nominal_freq
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_MANUFACTURER_NAME=IFD.property.dataobject.fairspec.nmr.instr_manufacturer_name
IFD_PROPERTY_DATAOBJECT_FAIRSPEC_NMR_INSTR_PROBE_TYPE=IFD.property.dataobject.fairspec.nmr.instr_probe_type

[# representations](#)

IFD_REP_DATAOBJECT_FAIRSPEC_HRMS_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.hrms.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_HRMS_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.hrms.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_HRMS_VENDOR_DATASET=IFD.representation.dataobject.fairspec.hrms.vendor_dataset

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IFD_REP_DATAOBJECT_FAIRSPEC_IR_JCAMP=IFD.representation.dataobject.fairspec.ir.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.ir.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.ir.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.ir.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_IR_SPECTRUM_TEXT=IFD.representation.dataobject.fairspec.ir.spectrum_text
IFD_REP_DATAOBJECT_FAIRSPEC_IR_VENDOR_DATASET=IFD.representation.dataobject.fairspec.ir.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_MS_JCAMP=IFD.representation.dataobject.fairspec.ms.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.ms.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.ms.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.ms.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_MS_SPECTRUM_TEXT=IFD.representation.dataobject.fairspec.ms.spectrum_text
IFD_REP_DATAOBJECT_FAIRSPEC_MS_VENDOR_DATASET=IFD.representation.dataobject.fairspec.ms.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_FID_1D=IFD.representation.dataobject.fairspec.nmr.jcamp_fid_1d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_FID_2D=IFD.representation.dataobject.fairspec.nmr.jcamp_fid_2d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_1i1r_1D=IFD.representation.dataobject.fairspec.nmr.jcamp_1i1r_1d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_1r_1D=IFD.representation.dataobject.fairspec.nmr.jcamp_1r_1d
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_JCAMP_2D=IFD.representation.dataobject.fairspec.nmr.jcamp_2d

IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.nmr.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_TEXT=IFD.representation.dataobject.fairspec.nmr.spectrum_text
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.nmr.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.nmr.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_NMR_VENDOR_DATASET=IFD.representation.dataobject.fairspec.nmr.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_JCAMP=IFD.representation.dataobject.fairspec.raman.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.raman.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.raman.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.raman.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_RAMAN_VENDOR_DATASET=IFD.representation.dataobject.fairspec.raman.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_JCAMP=IFD.representation.dataobject.fairspec.uvvis.jcamp
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_PEAKLIST=IFD.representation.dataobject.fairspec.uvvis.spectrum_peaklist
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_DESCRIPTION=IFD.representation.dataobject.fairspec.uvvis.spectrum_description
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_IMAGE=IFD.representation.dataobject.fairspec.uvvis.spectrum_image
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_SPECTRUM_DOCUMENT=IFD.representation.dataobject.fairspec.uvvis.spectrum_document
IFD_REP_DATAOBJECT_FAIRSPEC_UVVIS_VENDOR_DATASET=IFD.representation.dataobject.fairspec.uvvis.vendor_dataset

IFD_REP_DATAOBJECT_FAIRSPEC_COMP_INPUT=IFD.representation.dataobject.fairspec.comp.input
IFD_REP_DATAOBJECT_FAIRSPEC_COMP_OUTPUT=IFD.representation.dataobject.fairspec.comp.output

IFD_REP_DATAOBJECT_FAIRSPEC_XRAY_CIF=IFD.representation.dataobject.fairspec.xray.cif